Machining with Grinding Wheels

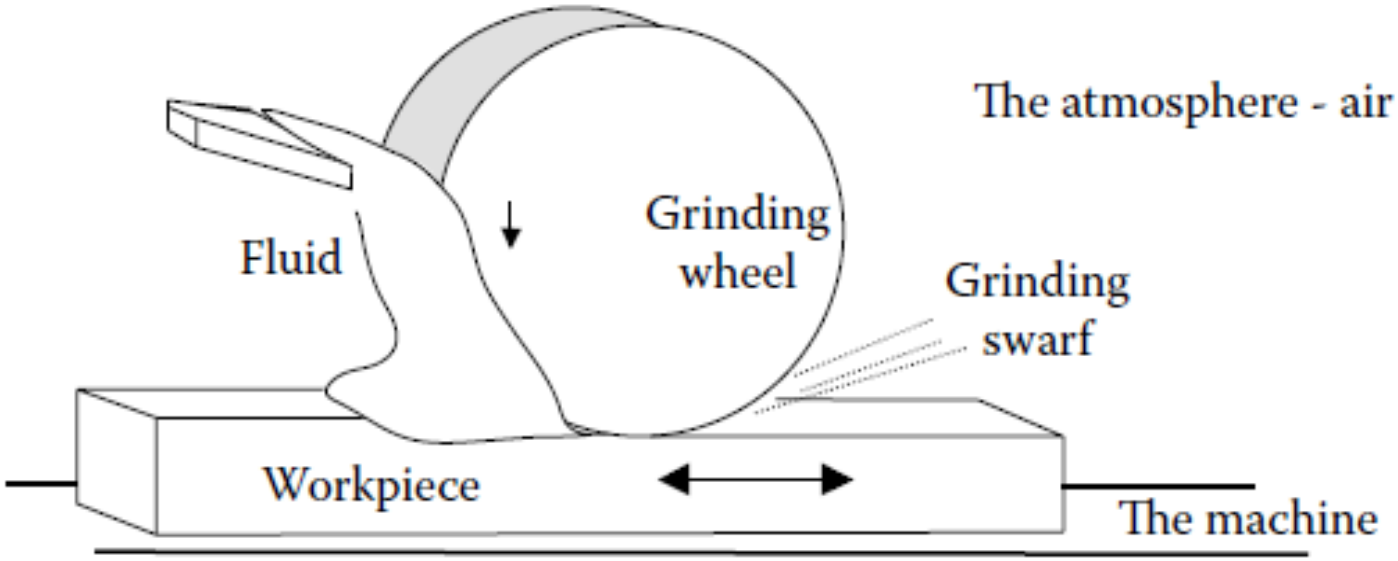
Abrasive Grinding Wheel Presentation

1. Grinding Wheels (3 Element 5 Factors)
2. The Basic Grinding Process
3. Marking System

BASIC USES OF GRINDING

- HIGH ACCURACY REQUIRED
  - สำหรับอัตราการแม่นยำสูง
- HIGH REMOVAL RATE REQUIRED
  - สำหรับอัตราการกำจัดสูง
- MACHINING OF HARD MATER
  - วัสดุที่แข็งและมีพื้นผิวที่แข็งมากๆ ในหลายๆ กรณีการเจียรอาจเป็นเพียงวิธีการเดียวในการจัดการผิวของชิ้นงาน

THE BASIC GRINDING PROCESS



The six basic elements involved in surface grinding.

6 Basic Elements in Grinding wheel

THE SIX BASIC ELEMENTS

Grinding wheel (หินเจียร)

- Abrasive (สารขัดถู), Bone (ตัวประสาน), Pore (โพรงอากาศ)

Work piece (ชิ้นงาน)

- Material (วัตถุดิบ), Size (ขนาด), Hardness (ความแข็ง), Removal (เจียรออก), Surface roughness (ความเรียบผิว)

The machine (เครื่องจักร)

- Grinding Operation (กระบวนการเจียร), Speed (ความเร็วรอบ),

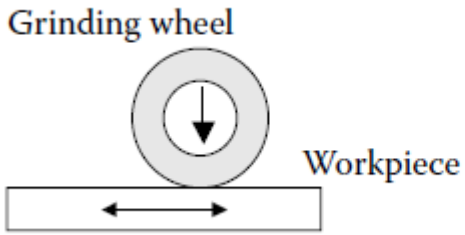
Fluid (ของเหลว)

- Coolant (สารหล่อเย็น), Wet or Dry

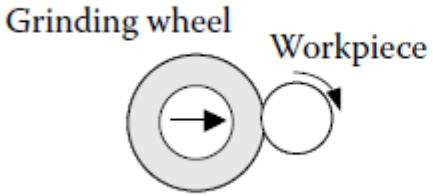
Grinding swarf (ซากเจียร)

The atmosphere – air (บรรยากาศ)

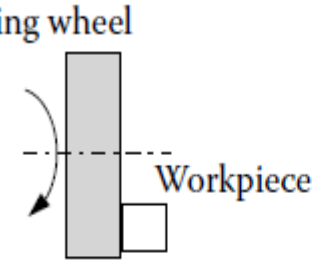
FOUR BASIC GRINDING OPERATIONS



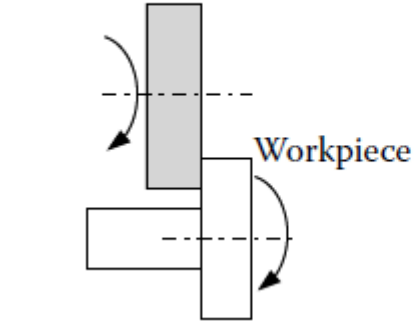
(a) Peripheral surface grinding



(b) Peripheral cylindrical grinding

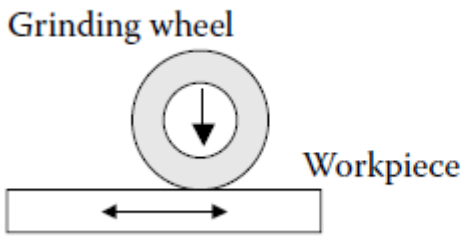


(c) Face surface grinding



(d) Face cylindrical grinding

PERIPHERAL SURFACE GRINDING



(a) Peripheral surface grinding

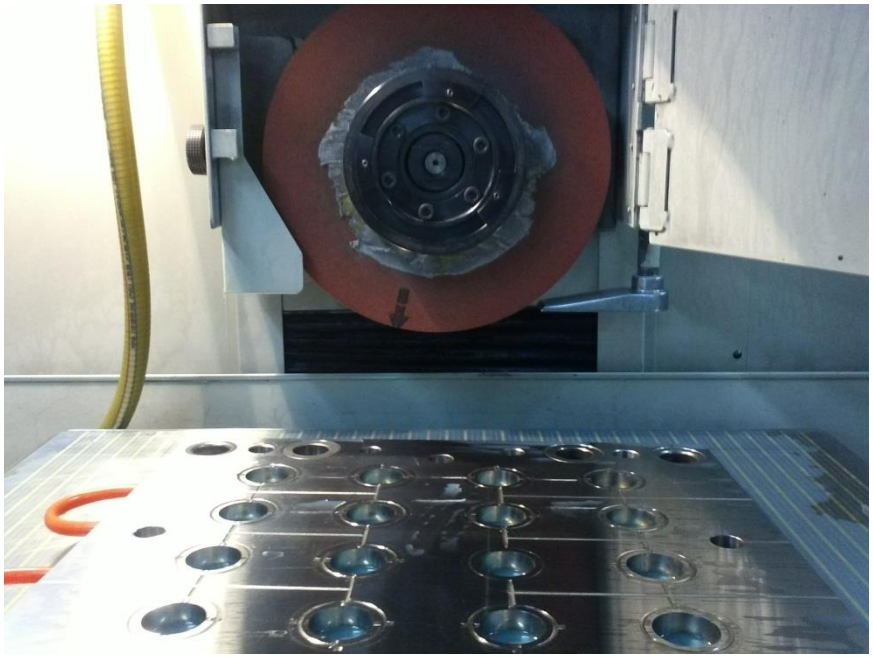
ข้อสังเกต

หินเจียรกลม

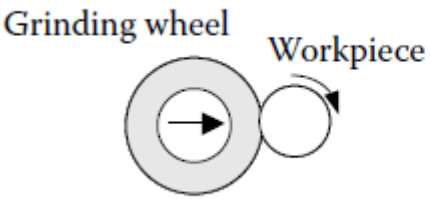
ชิ้นงานแบน



Surface Grinding
Snagging



PERIPHERAL CYLINDRICAL GRINDING



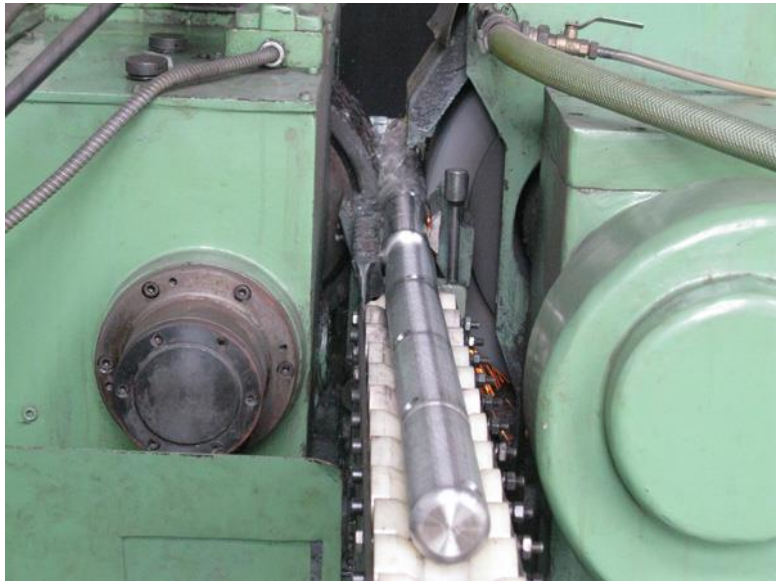
(b) Peripheral cylindrical grinding

ข้อสังเกต
หินเจียรกลม
ชิ้นงานกลม

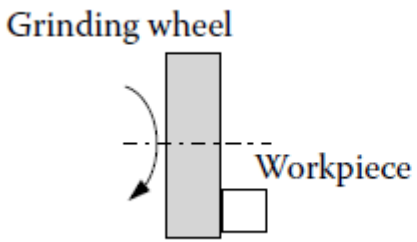
Cylindrical grinding
Center less
Internal







FACE SURFACE GRINDING



(c) Face surface grinding

ข้อสังเกต

หินเจียรไม่ว่าจะกลมหรือเป็นแท่งเหลี่ยมๆ จะใช้ด้านที่เรียบเจียรเสมอ

ชิ้นงานแบนราบ

Segment
Rotary
Double Disk

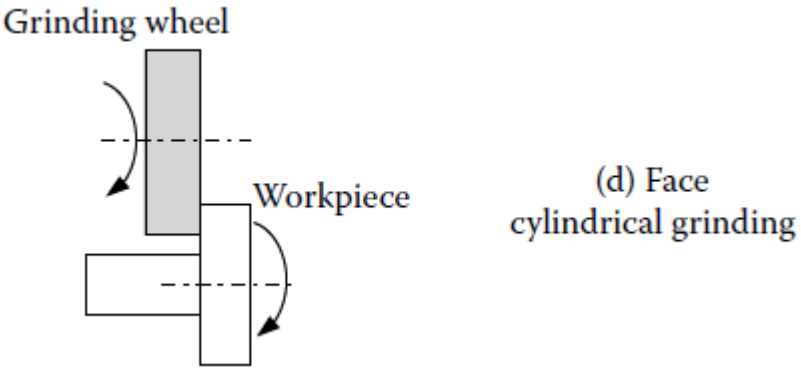
หินถ้วย 6A, เจียรฟันเลื่อย







FACE CYLINDRICAL GRINDING



ข้อสังเกต

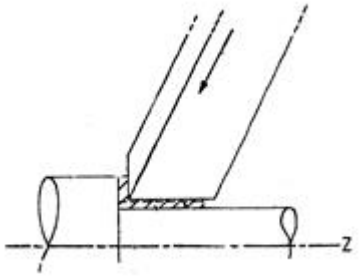
หินเจียรกลม ใช้ด้านข้างมุมของหิน เจียรชิ้นงาน
ชิ้นงานมักจะกลมและมีบ่าที่ต้องเจียร

Angular grinding
หินถ้วย 11A,12A









THE IMPORTANCE OF THE ABRASIVE

TABLE 1.1
Typical Hardness of Abrasive Grain Materials at Ambient Temperatures

	Units (GPa)
Diamond	56–102
Cubic boron nitride	42–46
Silicon carbide	~ 24
Aluminium oxide	~ 21

Source: From De Beers, 1983. With permission.

This means that the grain must be harder than the work piece at the temperature of the interaction.

If the work piece is harder than the grain, it is the grain that suffers most wear.

GRINDING WHEELS FOR A PURPOSE

- Grinding wheels vary enormously in design according to the purpose for which the wheel is to be used.
- There is scope for engineering bond properties to achieve strength and wear behavior suited to the particular abrasive within each class of bond.
- A general-purpose wheel will give greatly inferior removal rates and economics compared to an optimized wheel for the particular product.

PROBLEM-SOLVING

Few readers have time and fortitude to read a handbook from beginning to end. Although much could be learned from such an approach, readers are encouraged to cherry-pick their way through the most appropriate chapters.

PART I

- The 12 chapters in Part I cover the principles of grinding. This part includes all aspects that relate to grinding generally. Topics include basic grinding parameters, grinding wheels and grinding wheel structure, and wheel-dressing processes used for preparing wheels for grinding and used for restoring grinding efficiency.

PART II

- The 8 chapters in Part II explore applications of grinding. Part II covers grinding of conventional ductile materials, grinding of brittle-hard materials, grinding machine technology and rotary dressers, surface grinding, external cylindrical grinding, internal cylindrical grinding, centerless grinding, and ultrasonically assisted grinding. A particular emphasis is placed on developments in technology that can lead to improved part quality, higher productivity, and lower costs.

Chapter 2 Grinding Parameters

- **INTRODUCTION**
  - WHEEL LIFE
  - REDRESS LIFE
  - CYCLE TIME

PROCESS PARAMETERS

- UNCUT CHIP THICKNESS OR GRAIN PENETRATION DEPTH
- WHEEL SPEED
- WORK SPEED
- DEPTH OF CUT
- EQUIVALENT WHEEL DIAMETER
- ACTIVE GRIT DENSITY
- GRIT SHAPE FACTOR
- FORCE PER GRIT
- SPECIFIC GRINDING ENERGY
- SPECIFIC REMOVAL RATE
- GRINDING POWER
- TANGENTIAL GRINDING FORCE
- NORMAL GRINDING FORCE
- COEFFICIENT OF GRINDING

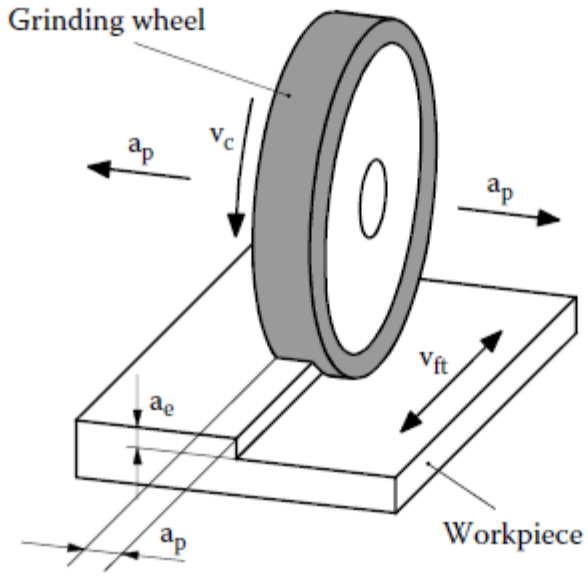
PROCESS PARAMETERS

- SURFACE ROUGHNESS
- *RT* ROUGHNESS
- *RA* ROUGHNESS
- *RZ* ROUGHNESS
- MATERIAL OR BEARING RATIO
- PEAK COUNT
- COMPARISON OF ROUGHNESS CLASSES
- FACTORS THAT AFFECT ROUGHNESS MEASUREMENTS
- ROUGHNESS SPECIFICATIONS ON DRAWINGS
- STOCK REMOVAL PARAMETER
- DECAY CONSTANT τ
- G-RATIO
- P-RATIO
- CONTACT LENGTH
- GEOMETRIC CONTACT LENGTH
- REAL CONTACT LENGTH

GRINDING TEMPERATURES

- SURFACE TEMPERATURE *T*
- MAXIMUM WORKPIECE SURFACE TEMPERATURE
- THE Cmax FACTOR
- THE TRANSIENT THERMAL PROPERTY β*W*
- WORKPIECE PARTITION RATIO *RW*
- EFFECT OF GRINDING VARIABLES ON TEMPERATURE
- HEAT CONVECTION BY COOLANT AND CHIPS
- CONTROL OF THERMAL DAMAGE

WHEEL SPEED, WORK SPEED



a_p = Feed in the direction of the grinding wheel

a_e = Feed relative the workpiece surface

v_{ft} = Feed speed

v_c = Cutting speed

FIGURE 16.3 Longitudinal peripheral grinding.

Parameter	Acceptable Symbols				
Grinding wheel speed	v_s	or	v_c		
Workspeed	v_w	or	v_{ft}		
Width of grinding	b_w	or	a_p	or	$b_{s.eff}$

WHEEL SPEED

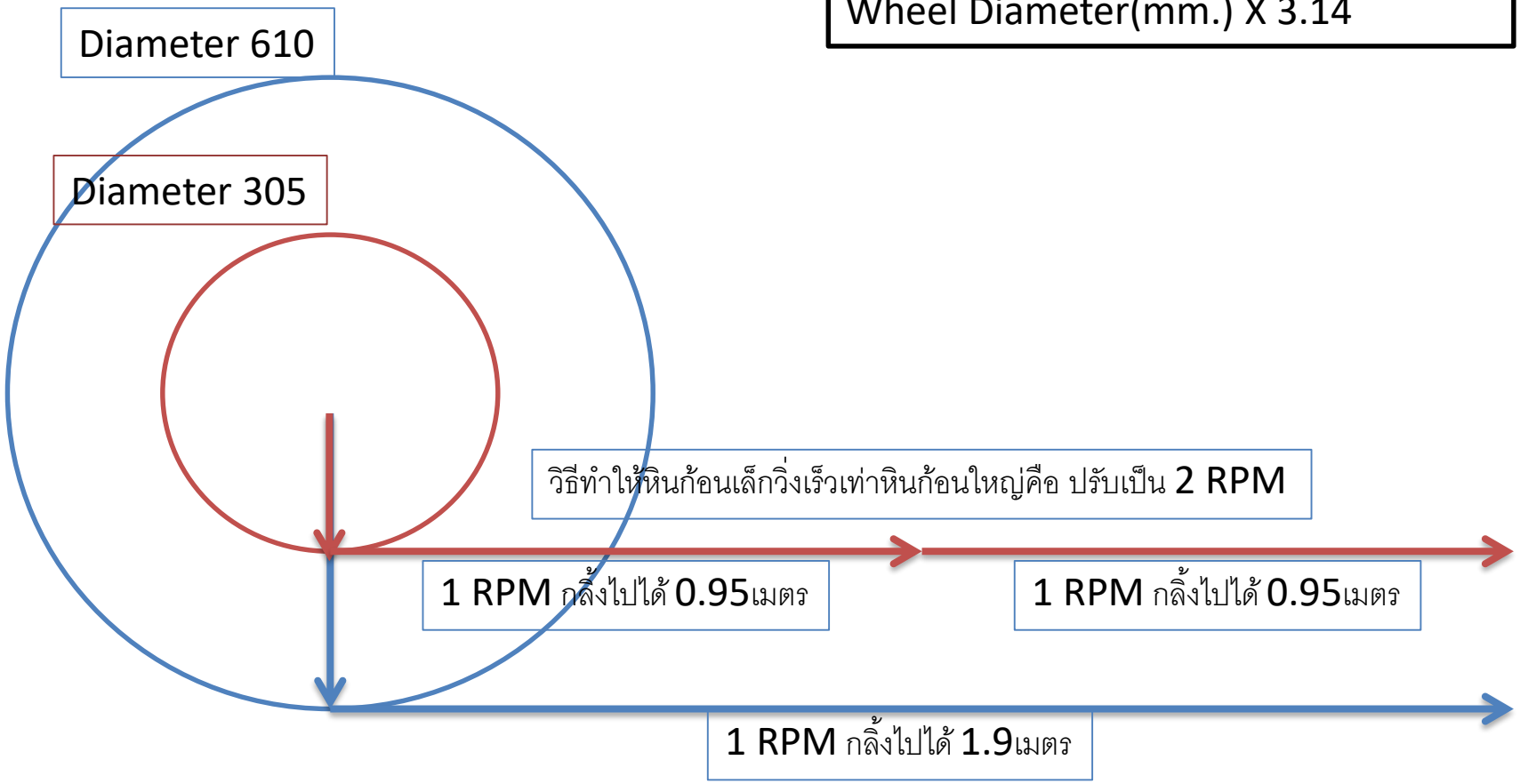
$$\frac{\text{cutting speed (M/S)} \times 60{,}000}{\text{wheel diameter (mm)} \times 3.14} = \text{R.P.M.}$$

$$\frac{\text{R.P.M.} \times \text{wheel diameter (mm)} \times 3.14}{60{,}000} = \text{cutting speed (M/S)}$$

Inch	Minlimeter	25	30	35	45	50
1	25	19,091	22,909	26,727	34,364	38,182
6	150	3,182	3,818	4,455	5,727	6,364
12	300	1,591	1,909	2,227	2,864	3,182
14	350	1,364	1,636	1,909	2,455	2,727
16	400	1,193	1,432	1,670	2,148	2,386
18	450	1,061	1,273	1,485	1,909	2,121
20	500	955	1,145	1,336	1,718	1,909
30	750	636	764	891	1,145	1,273

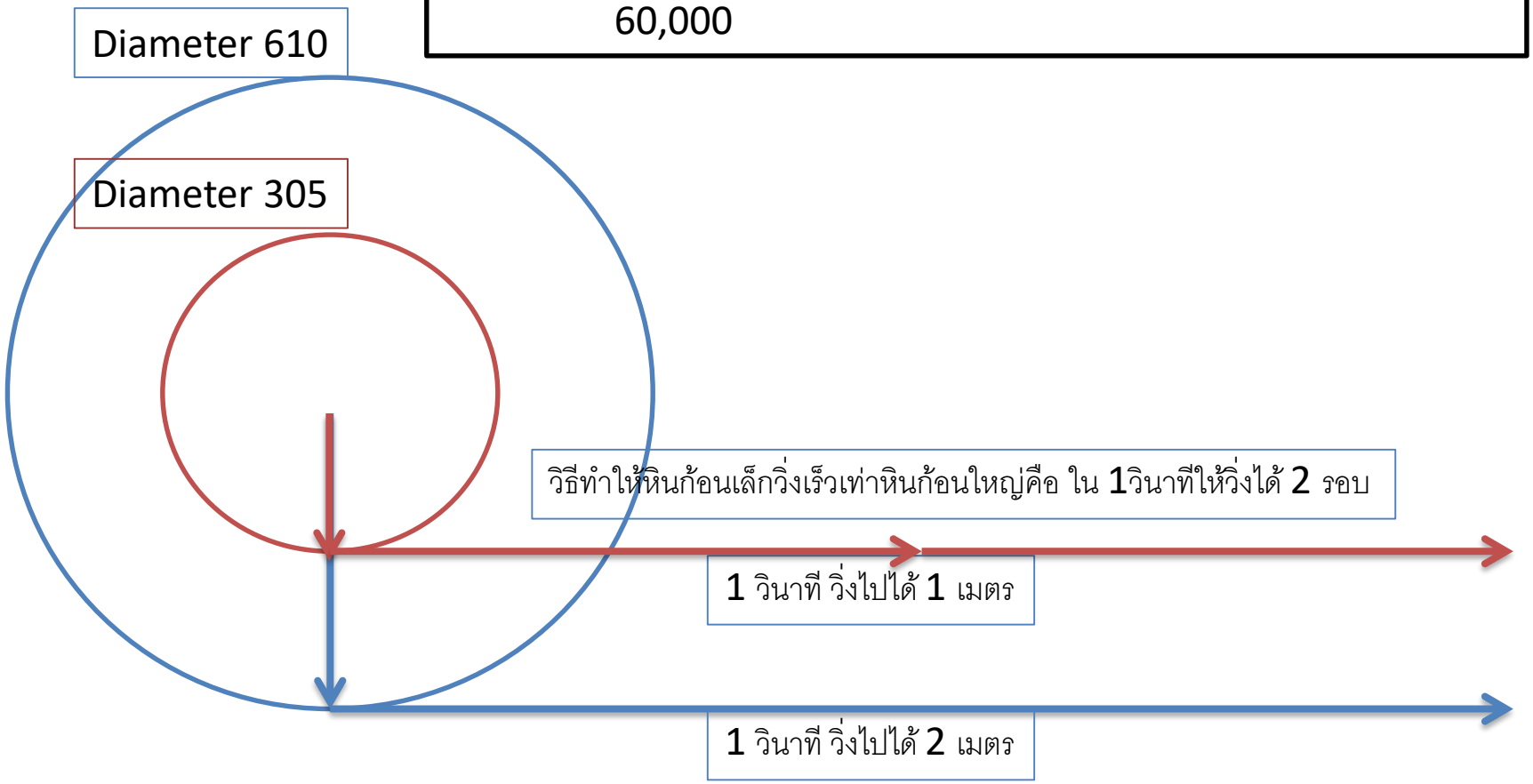
RPM

$$\frac{\text{Cutting Speed(M/S) X 60,000}}{\text{Wheel Diameter(mm.) X 3.14}} = \text{RPM}$$



M/S (เมตร/นาที)

$$\frac{\text{RPM X Wheel Diameter(mm.) X 3.14}}{60{,}000} = \text{Cutting Speed(M/S)}$$



DRAWING FORM

APPENDIX 2.1 DRAWING FORM AND PROFILE TOLERANCING

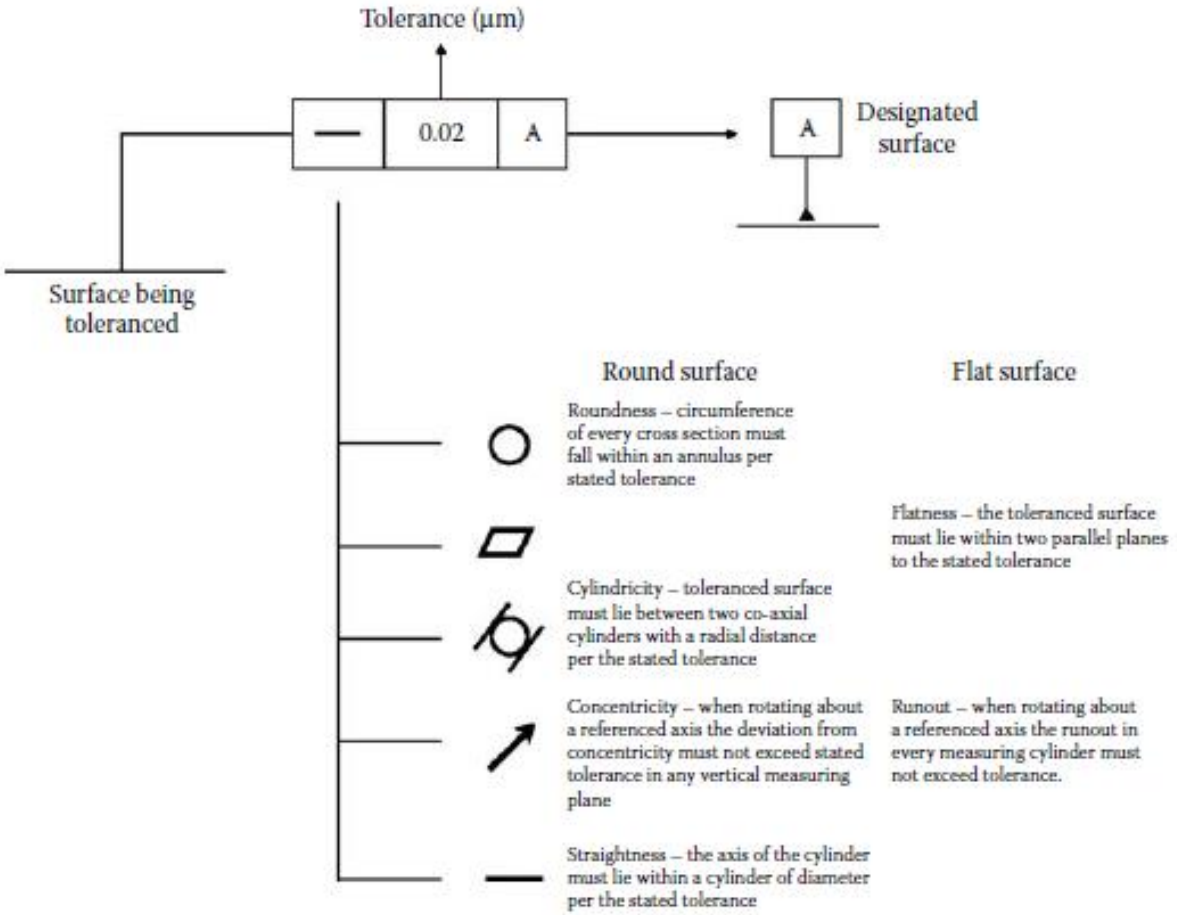
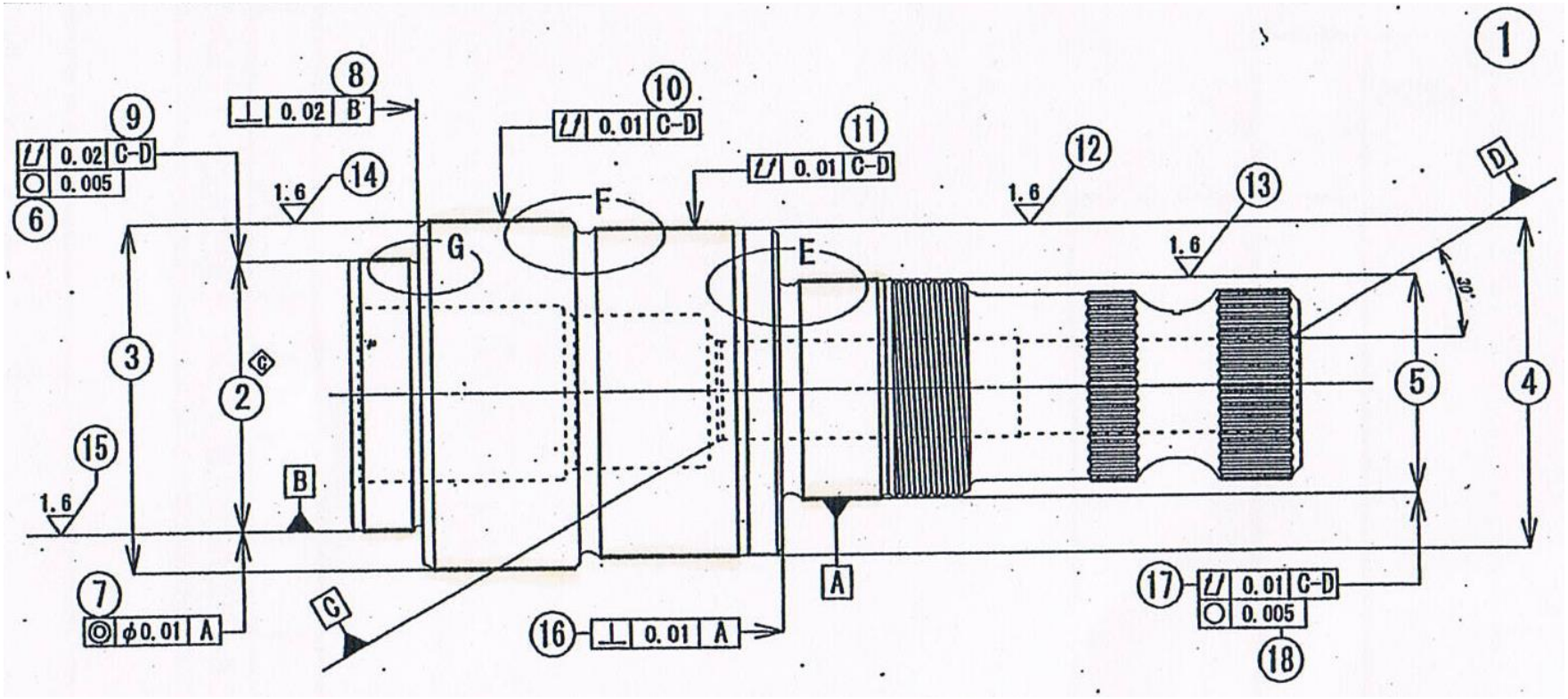
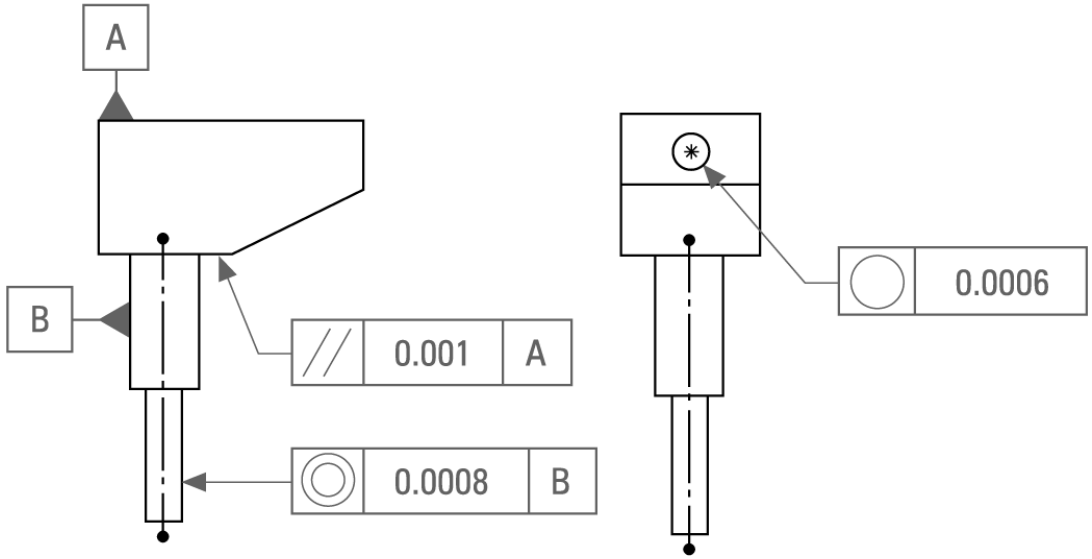
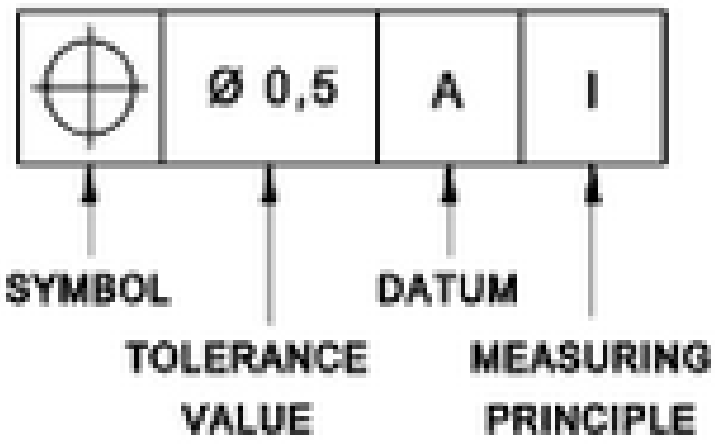


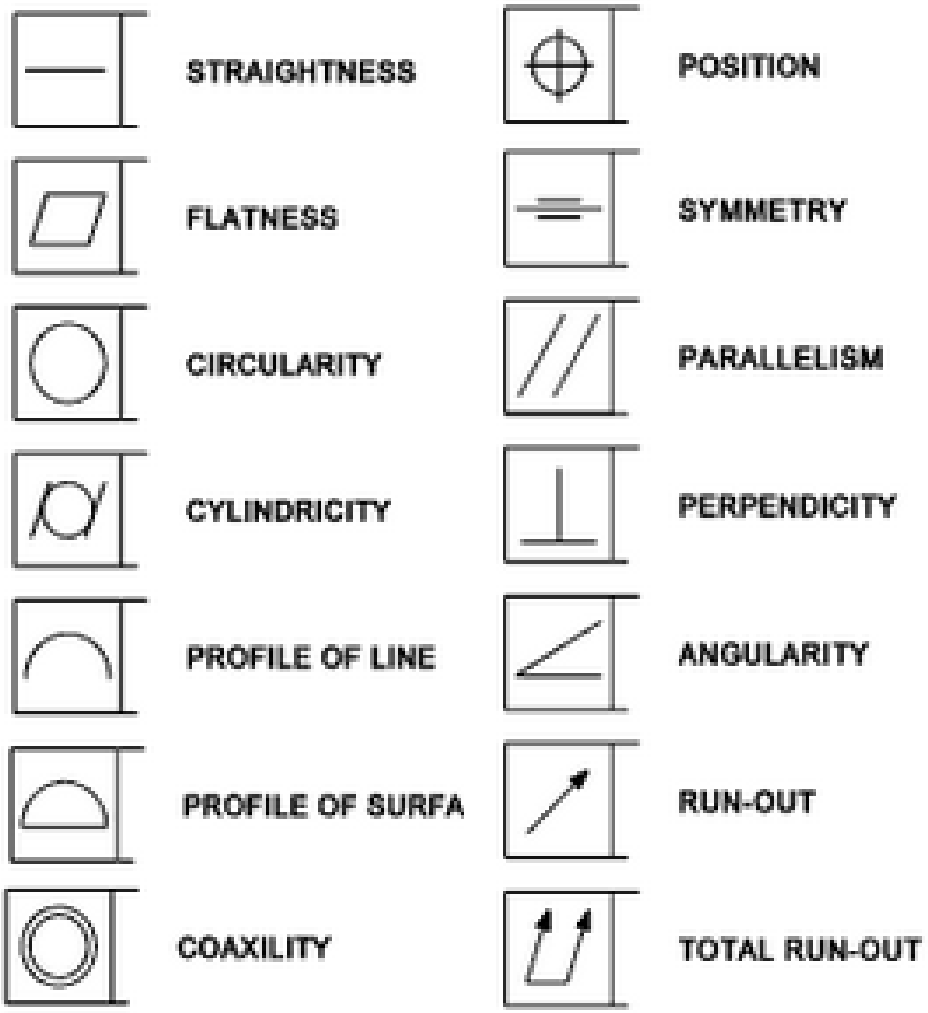
FIGURE A2.1 Examples of form and position tolerances.


1
8
⊥ 0.02 B
9
// 0.02 C-D
○ 0.005
6
10
// 0.01 C-D
11
// 0.01 C-D
12
1.6
13
1.6
D
14
1.6
F
G
E
30°
3
2
C
5
4
15
1.6
B
A
7
◎ φ0.01 A
C
16
⊥ 0.01 A
17
// 0.01 C-D
○ 0.005
18

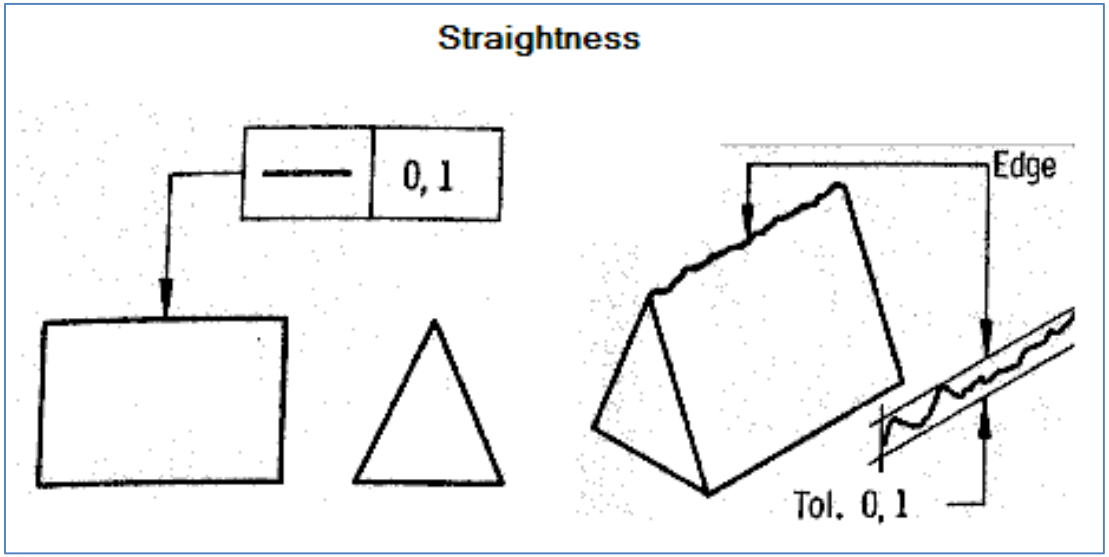

A
B
// 0.001 A
◎ 0.0008 B
✳
○ 0.0006

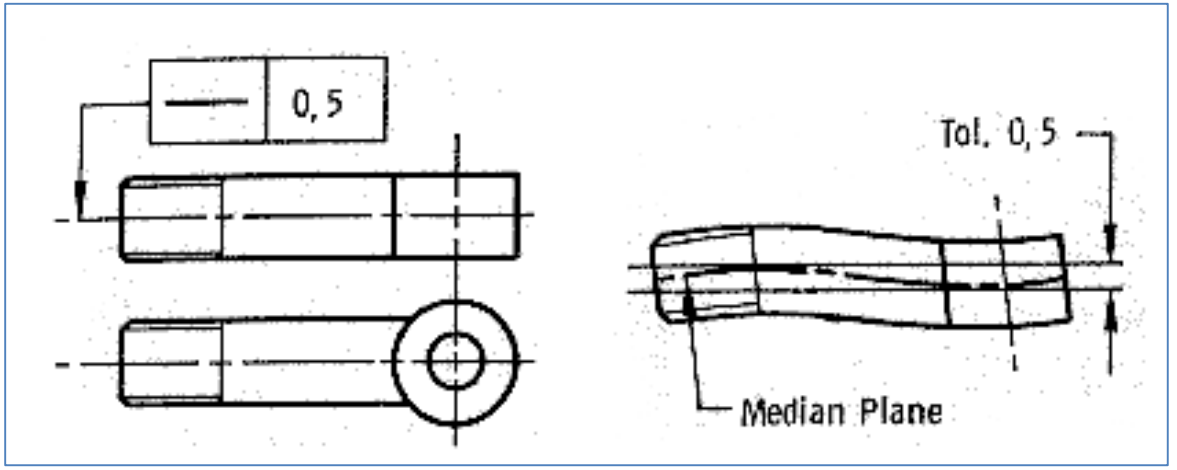
SPECIFICATIONS ON DRAWINGS



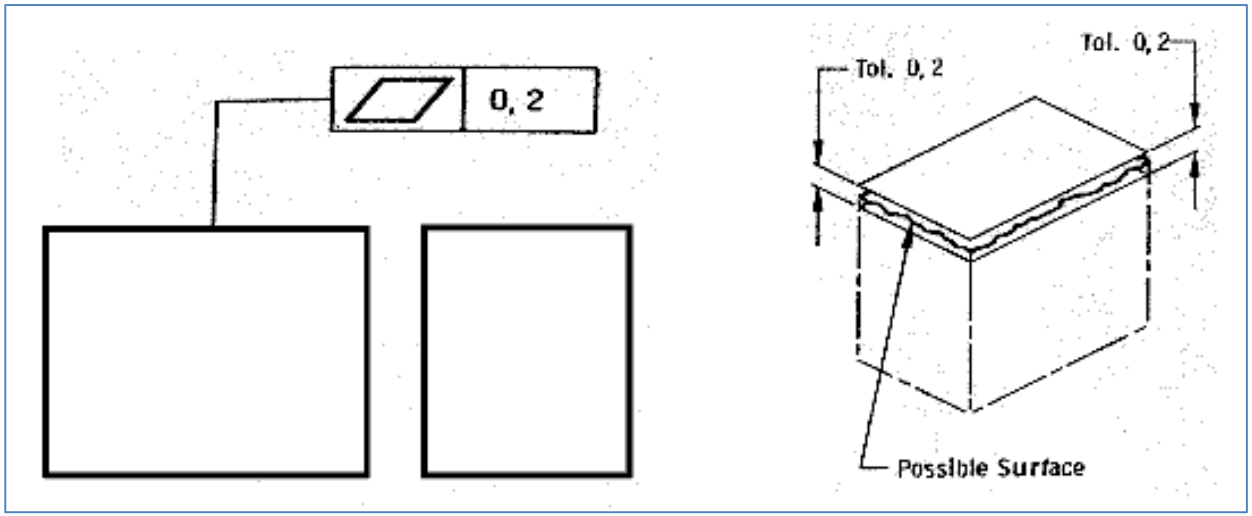


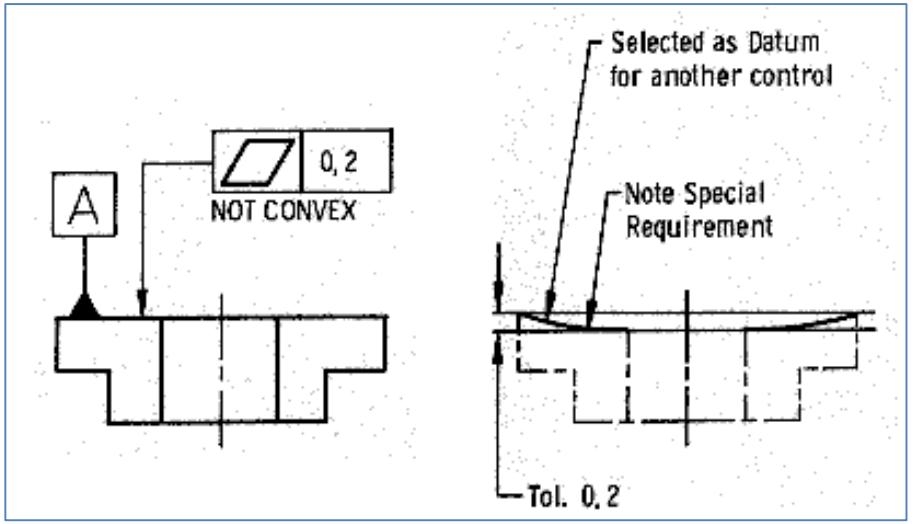
STRAIGHTNESS



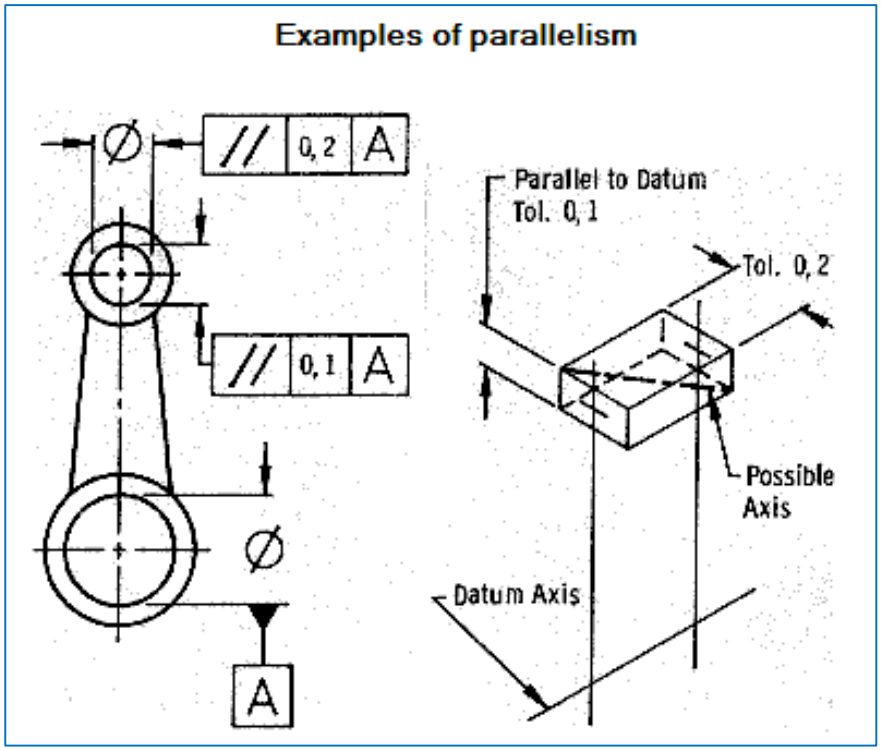


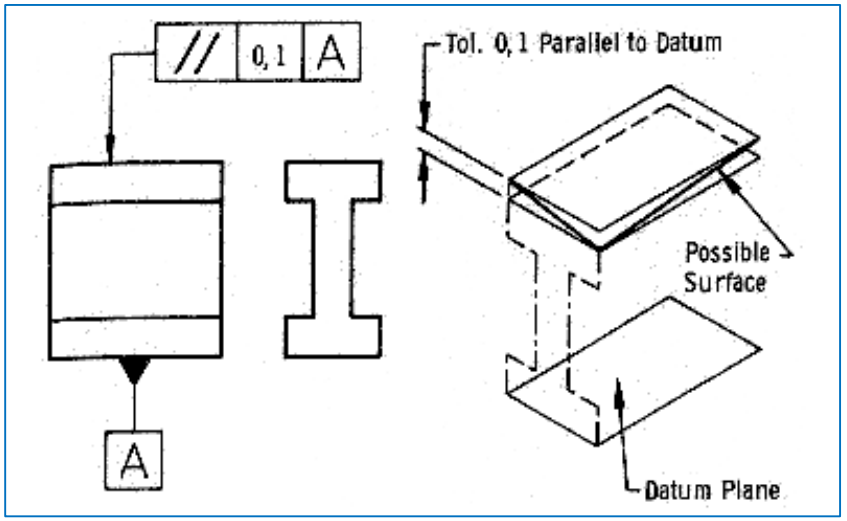
FLATNESS


0, 2
Tol. 0, 2
Tol. 0, 2
Possible Surface

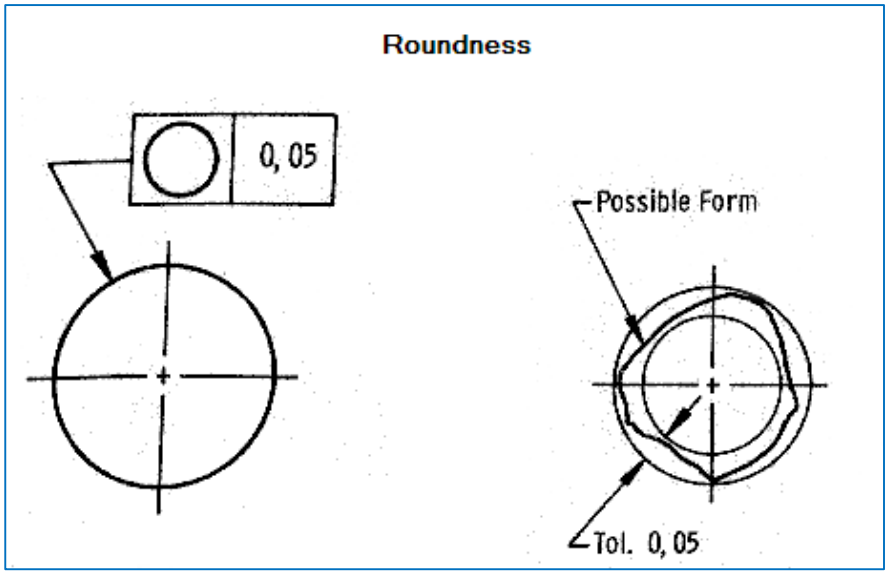

A
0, 2
NOT CONVEX
Selected as Datum
for another control
Note Special
Requirement
Tol. 0, 2

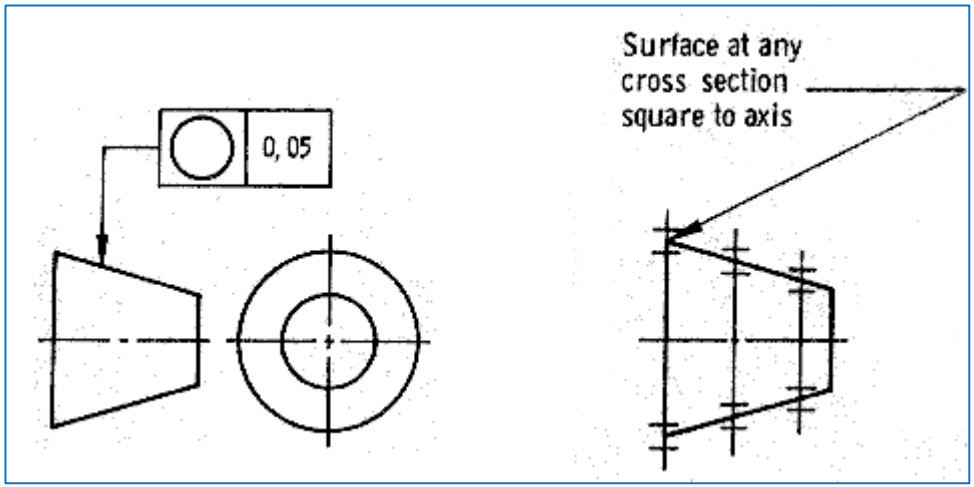
PARALLELISM



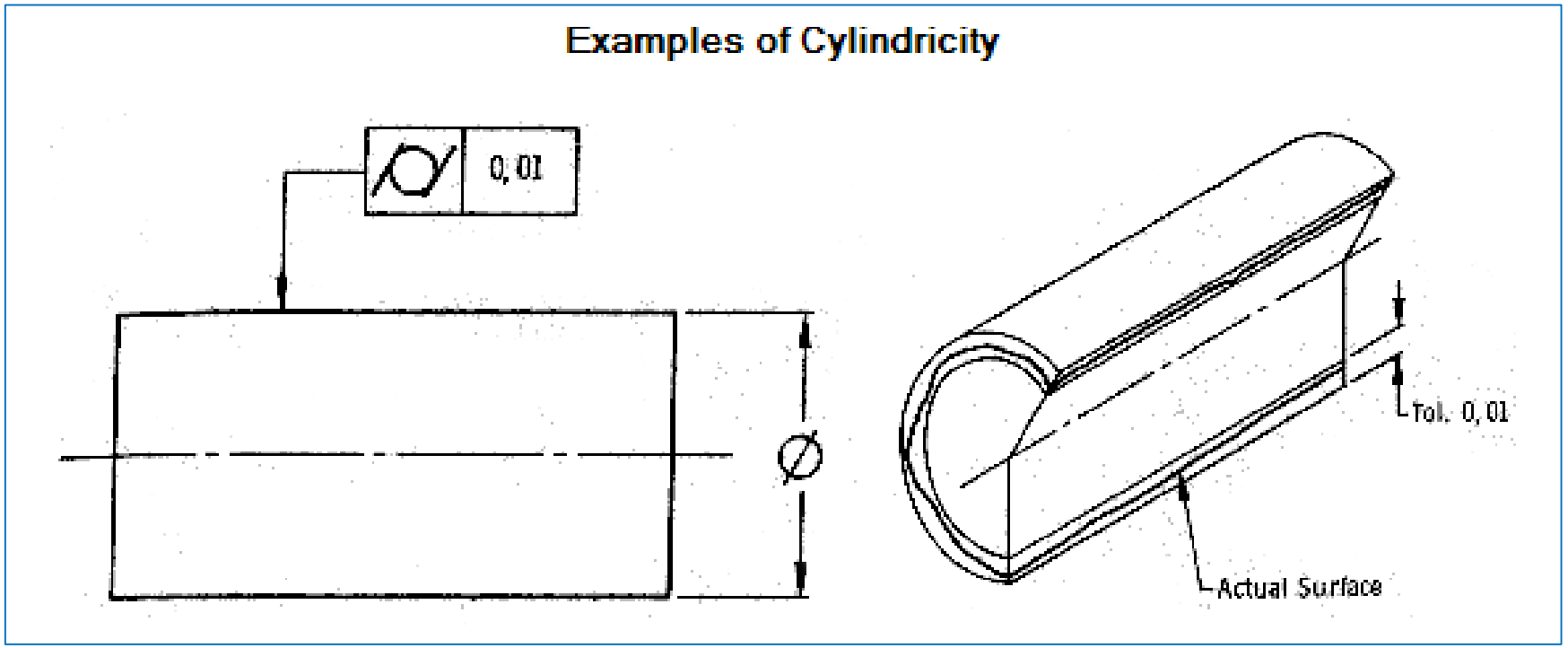


ROUNDNESS

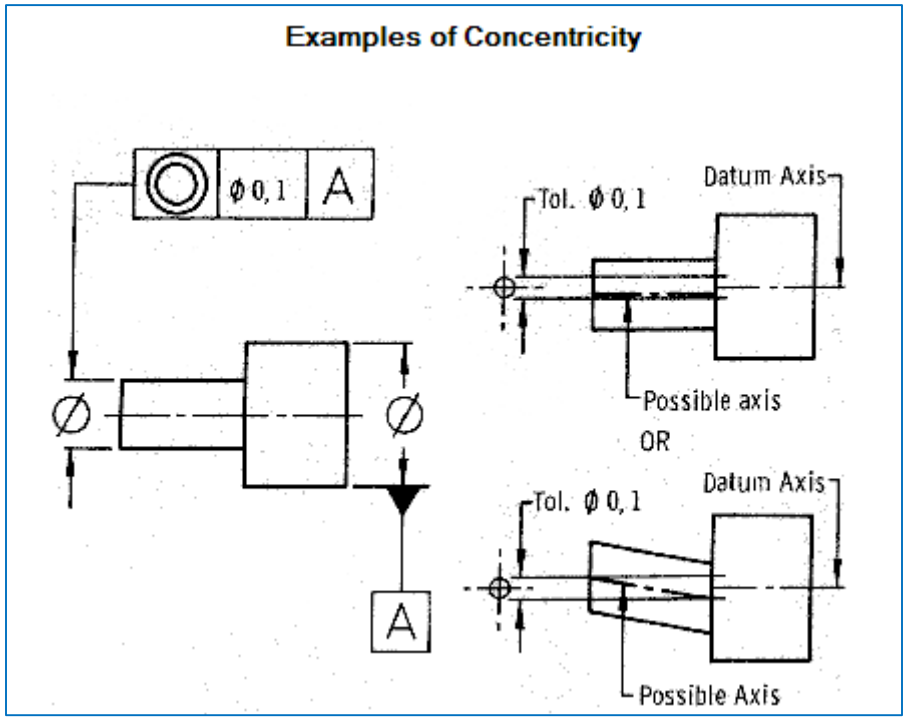

Roundness
0, 05
Possible Form
Tol. 0, 05

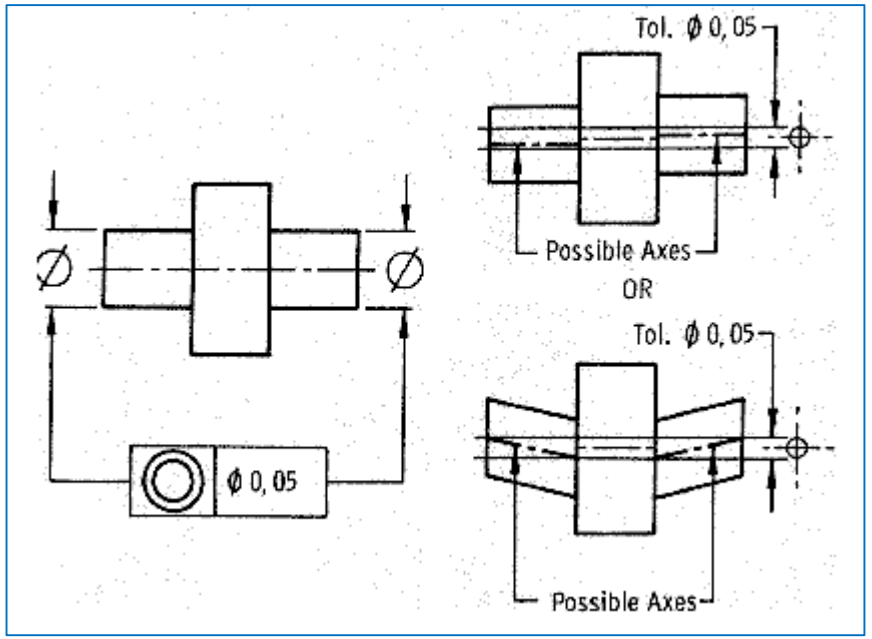

0, 05
Surface at any
cross section
square to axis

CYLINDRICITY

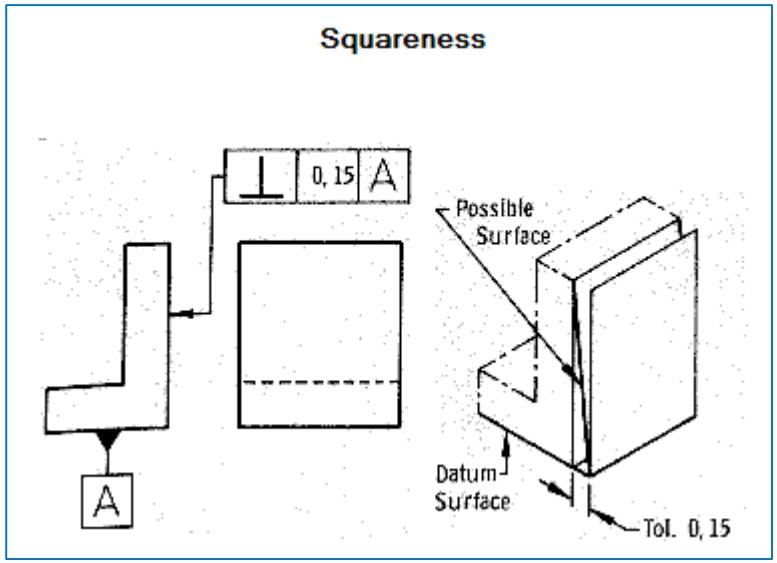


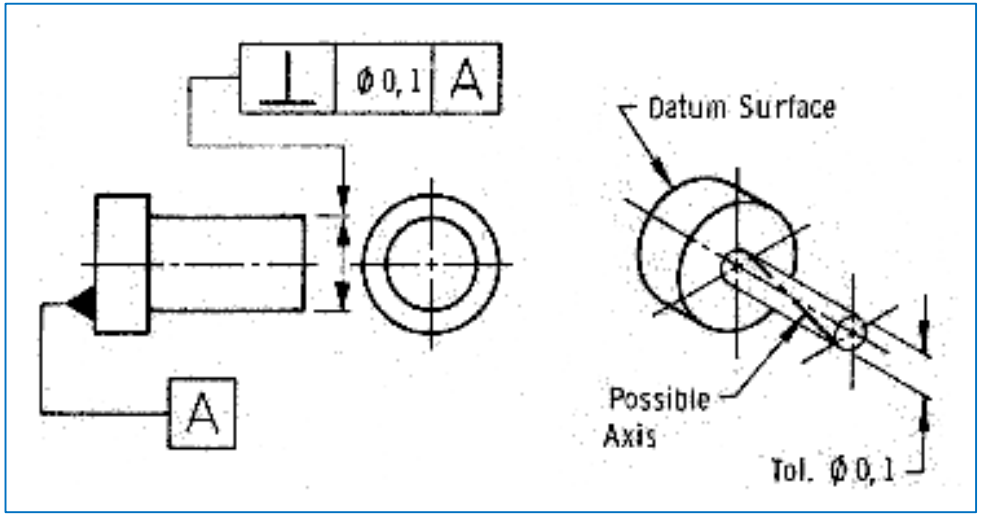
CONCENTRICITY



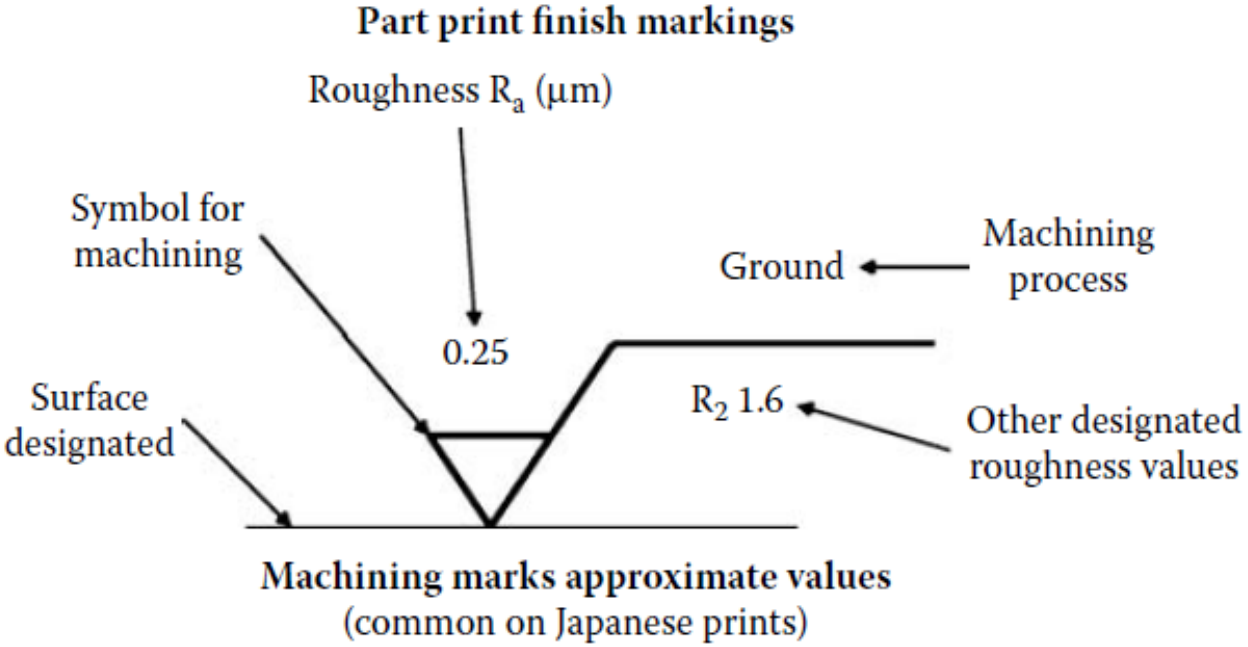


SQUARENESS


Squareness
0, 15
A
A
Possible
Surface
Datum
Surface
Tol. 0, 15


Ø 0, 1
A
A
Datum Surface
Possible
Axis
Tol. Ø 0, 1

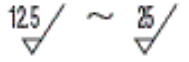
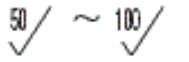
Roughness Specifications on Drawings



Print markings for surface finish.

Conventional finishing symbol

Reference: Relationship Between Arithmetic Average Roughness (Ra) and Previous Notation

Arithmetical average roughness Ra			Max. height Ry	Ten-spot average roughness Rz	Ry·Rz reference length ℓ (mm)	Conventional finishing symbol
Standard sequence	Cut-off value λc (mm)	Drawing indication of surface texture	Standard sequence			
0.012 a 0.025 a 0.05 a 0.1 a 0.2 a	0.08		0.05 s 0.1 s 0.2 s 0.4 s 0.8 s	0.05 z 0.1 z 0.2 z 0.4 z 0.8 z	0.08	
	0.25				0.08	
	0.25				0.25	
	0.8				0.25	
	0.8				0.8	
0.4 a 0.8 a 1.6 a	0.8		1.6 s 3.2 s 6.3 s	1.6 z 3.2 z 6.3 z	0.8	
3.2 a 6.3 a	2.5		12.5 s 25 s	12.5 z 25 z	2.5	
12.5 a 25 a	8		50 s 100 s	50 z 100 z	2.5	
					8	
50 a 100 a	8		200 s 400 s	200 z 400 z	8	~
	-				-	

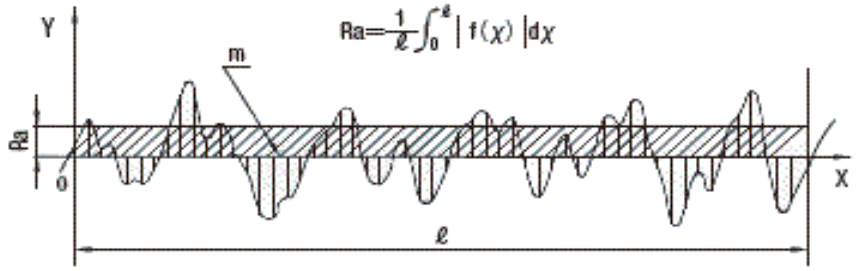
*The relationships among the three varieties shown here are not precise, and are presented for convenience only.

*Ra: The evaluation lengths of Ry and Rz are the cut-off values and the reference length each multiplied by five.

SURFACE ROUGHNESS

Arithmetical average roughness (Ra)

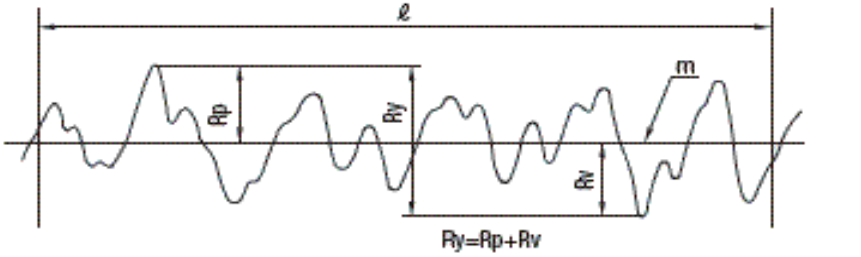
A portion stretching over a reference length in the direction in which the average line extends is cut out from the roughness curve. This portion is presented in a new graph with the X axis extending in the same direction as the average line and the Y axis representing the magnitude. When the roughness curve is represented by y=f (X), Ra is the value in microns (μm) found from the formula shown at right.



Maximum height (Ry)

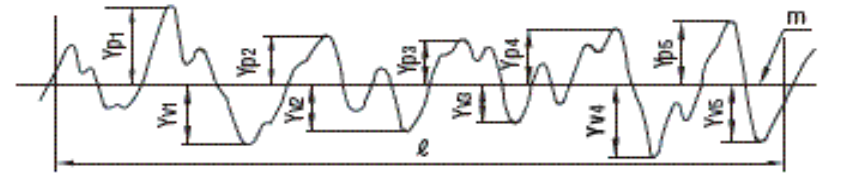
A portion stretching over a reference length in the direction in which the average line extends is cut out from the roughness curve. The gap between the peak line and valley line in this portion is measured in the direction of the magnitude axis, and this value is indicated in microns (μm).

Note: When finding Ry, the reference length is selected from a portion which contains no abnormally high peaks or abnormally low valleys (locations which are likely flaws).



Ten-spot average roughness (Rz)

A portion stretching over a reference length in the direction in which the average line extends is cut out from the roughness curve. Within this portion, the average absolute value of the height (Yp) of the five highest peaks as measured from the average line and the average absolute value of the height (Yv) of the five lowest valleys are added together. Rz is this sum, in microns (μm).

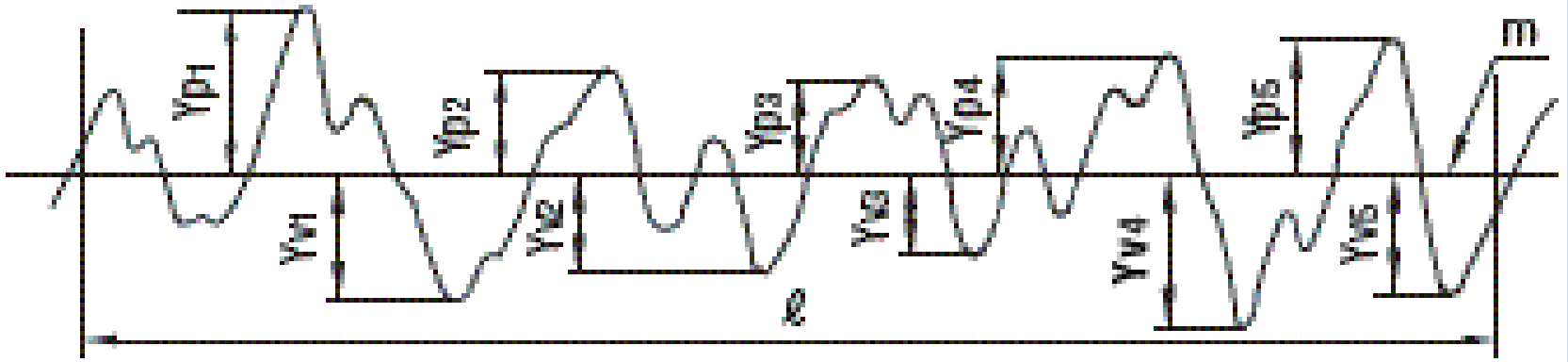


$$Rz = \frac{|Yp_1+Yp_2+Yp_3+Yp_4+Yp_5| + |Yv_1+Yv_2+Yv_3+Yv_4+Yv_5|}{5}$$

Yp_1、Yp_2、Yp_3、Yp_4、Yp_5 : Heights of the of top five peaks within the sampled portion of reference length ℓ

Yv_1、Yv_2、Yv_3、Yv_4、Yv_5 : Heights of the five lowest valleys within the sampled portion of reference length ℓ

Rz ROUGHNESS



$$Rz = \frac{\left| Yp_1 + Yp_2 + Yp_3 + Yp_4 + Yp_5 \right| + \left| Yv_1 + Yv_2 + Yv_3 + Yv_4 + Yv_5 \right|}{5}$$

Yp_1, Yp_2, Yp_3, Yp_4, Yp_5 : Heights of the of top five peaks within the sampled portion of reference length ℓ

Yv_1, Yv_2, Yv_3, Yv_4, Yv_5 : Heights of the five lowest valleys within the sampled portion of reference length ℓ

How to choose grit no. by require roughness

Grit No	Roughness Parameter Rz(µm)	Roughness Parameter Ra(µm)	Roughness Parameter Ry(µm)	Roughness Class	Roughness CLA (µ inch)
	12.8	3.2	25	N8	128
46	6.4	1.6	12.8	N7	64
	4	1	8		40
60	3.2	0.8	6.4	N6	32
	2.5	0.63	5		25.2
80	2	0.5	4		20
80/100	1.6	0.4	3.2	N5	16
100	1.25	0.35	2.5		12.6
120	1	0.25	2		10
120/150	0.8	0.2	1.6	N4	8
	0.4	0.1	0.8	N3	4
200	0.2	0.05	0.4	N2	2
	0.1	0.025	0.2	N1	1

Comparisons of ISFS

TABLE 2.1
Guideline Comparisons of International Surface Finish Systems

Ra (μm)	Rt (μm)	Rz (μm)	RMS (μin.)	CLA (μin.)	PVA (μin.)	Roughness Class France (μm)	Renault R France (μm)	Citreon b France (μm)	Citreon V France (μm)	Roughness Class China	Quality Class Russia
0.025	0.2	0.16	1.12	1	6.3	12C	0.13	0.08	0.15	N1	12
0.05	0.3	0.32	2.2	2	12	11C	0.25	0.15	0.3	N2	11
0.06	0.5	0.38	2.7	2.4	16	11B	0.3	0.18	0.36	N2	11
0.08	0.6	0.5	3.6	3.2	20	11A	0.4	0.24	0.48	N2/N3	11
0.1	0.8	0.63	4.5	4	25	10C	0.5	0.3	0.6	N3	10
0.12	1	0.75	5.3	5	32	10B	0.6	0.37	0.73	N3	10
0.16	1.25	1	7.1	6.3	40	10A	0.8	0.48	0.97	N3/N4	10
0.2	1.5	1.25	9	8	50	9C	1	0.61	1.22	N4	9
0.25	2	1.6	11.2	10	63	9B	1.25	0.76	1.52	N4	9
0.31	2.5	2	14	12.5	80	9A	1.6	0.95	1.8	N4/N5	9
0.4	3.2	2.5	18	16	100	8C	2	1.2	2.4	N5	8
0.5	4	3.2	22.4	20	125	8B	2.5	1.5	3	N5	8
0.63	5	4	28	25	160	8A	3.2	1.9	3.8	N5/N6	8
0.8	6.3	5	35.5	31.5	200	7C	4	2.4	4.8	N6	7
1	8	6.3	45	40	250	7B	5	3	6	N6	7
1.25	10	8	56	50	320	7A	6.3	3.8	7.6	N6/N7	7
1.6	12.5	10	71	63	400	6C	8	4.7	9.4	N7	

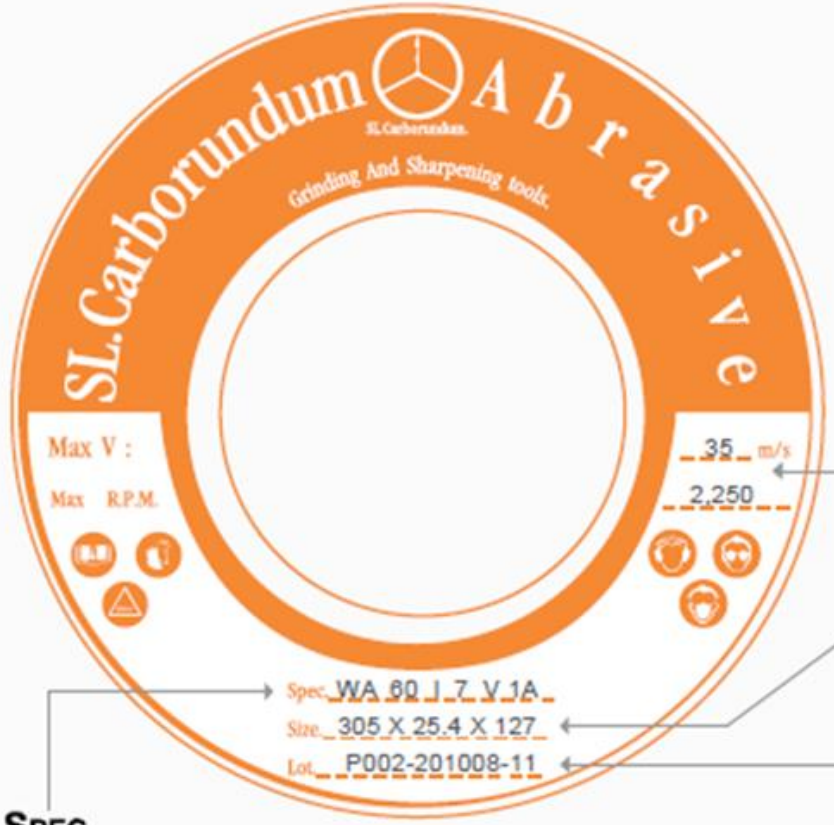
Chapter 3 Material Removal Mechanisms

- Significance
- Grinding Wheel Topography
- Determination of Grinding Wheel Topography
- Kinematics of the Cutting Edge Engagement
- Fundamental Removal Mechanisms
- Material Removal in Grinding of Ductile Materials
- Surface Formation in Grinding of Brittle-Hard Materials
- Energy Transformation

Chapter 4 Grinding Wheels

- DEVELOPMENTS IN PRODUCTIVITY
- SYSTEM DEVELOPMENT
- CONVENTIONAL AND SUPERABRASIVE WHEEL DESIGN

Marking System



WHEEL SPECIFICATION

Maximum safe operating speed

Wheel dimensions

305 X 25 X 127

DIAMETER THICHNESS HOLE

LOT.Number

SPEC

WA	60	I	7	V	1	A

ABRASIVE GRAIN SIZE GRADE STRUCTURE BOND TYPE FACE

WA60I7V1A 305X25X127

1.	GRAIN (สารขัดถู)			
2.	GRIT SIZE (ขนาดเม็ดเกรน)			
3.	GRADE (ความแข็ง)			
4.	STRUCTURE (โครงสร้าง)			
5.	KIND OF BOND (ประเภทของบอนด์)			
6.	SHAPE (รูปร่าง)			
7.	PROFILE (หน้าหิน)			
8.	Diameter	D	OD	(ความกว้าง)
9.	Thickness	T	T	(ความหนา)
10.	HOLE	H	ID	(รูใน)

- _WA
- _60
- _I
- _7
- _V
- _1
- _A
- _305
- _25
- _127

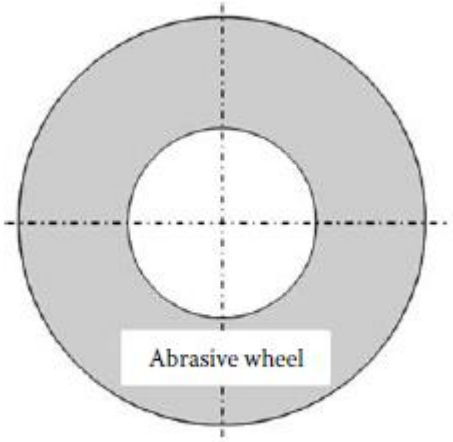
WHEEL SHAPE SPECIFICATION

- **BASIC SHAPES**

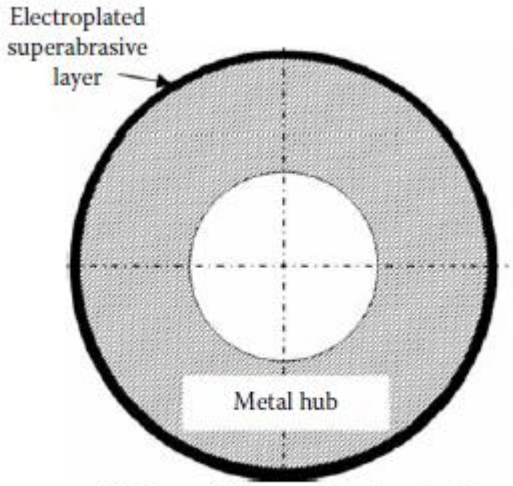
  Wheel dimensions are usually expressed as

  diameter(D) × thickness(T) × hole(H).

  For super abrasives the layer thickness(X) is added afterward.



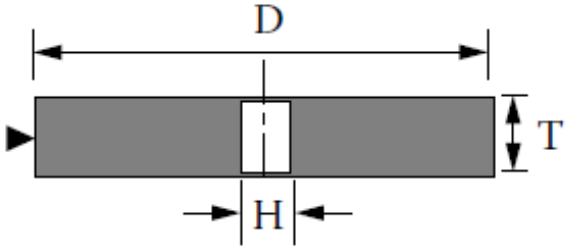
Low- or medium-speed conventional abrasive



High-speed EP superabrasive wheel

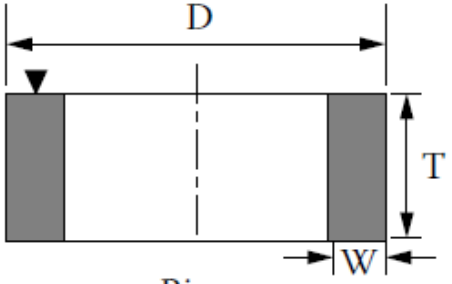
International Standard Shapes for Conventional Wheels

Type 1



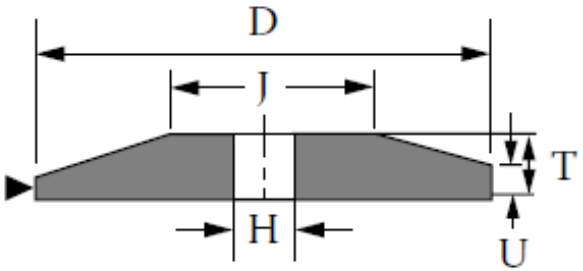
Plain

Type 2



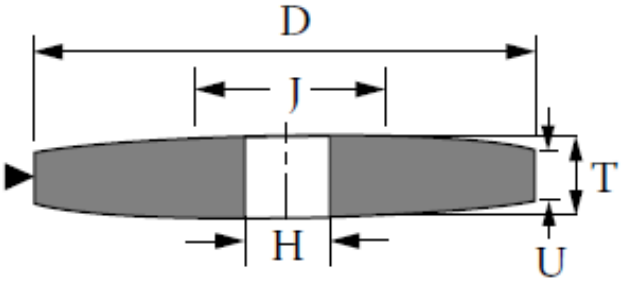
Ring

Type 3



Tapered one side

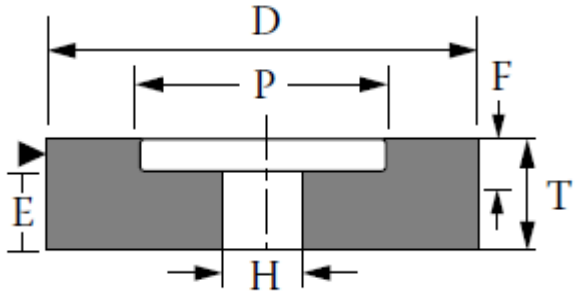
Type 4



Tapered two sides

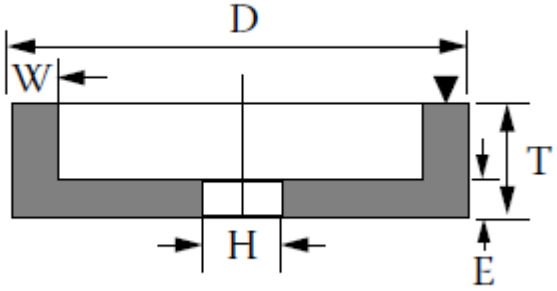
International Standard Shapes for Conventional Wheels

Type 5



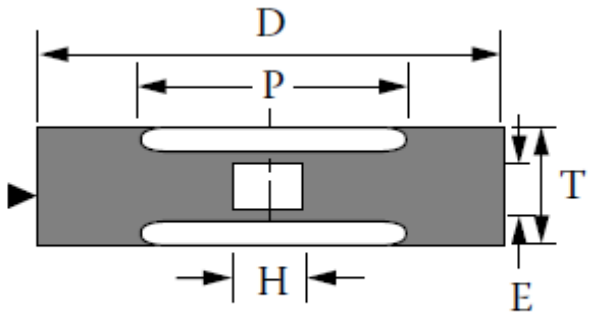
Recessed

Type 6



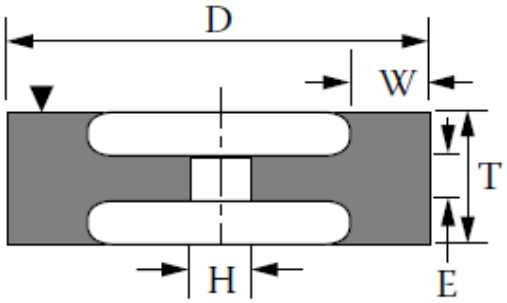
Straight cup

Type 7



Double recess

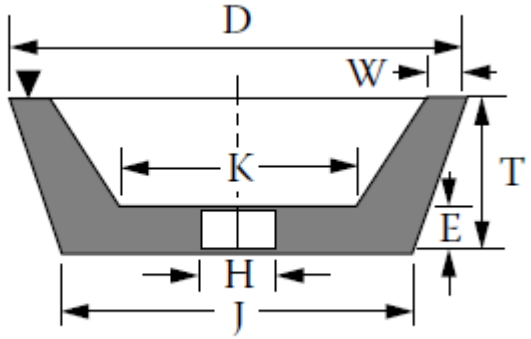
Type 9



Double cup

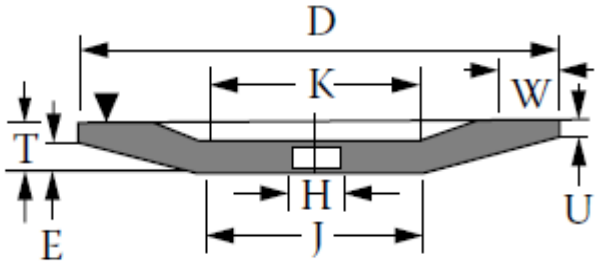
International Standard Shapes for Conventional Wheels

Type 11



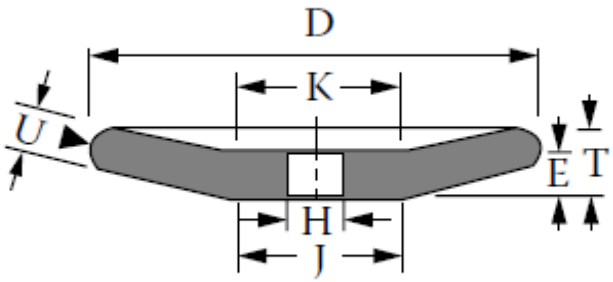
Taper cup

Type 12



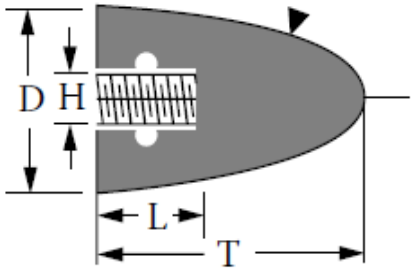
Dish

Type 13



Saucer

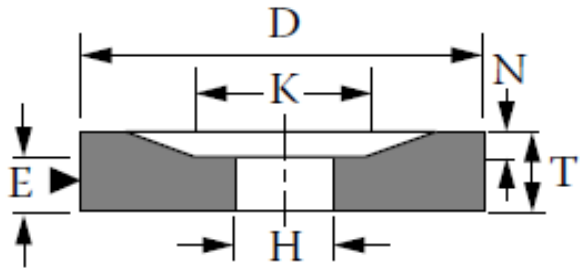
Type 16



Cone

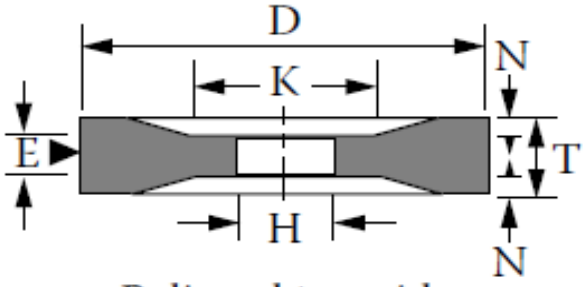
International Standard Shapes for Conventional Wheels

Type 20



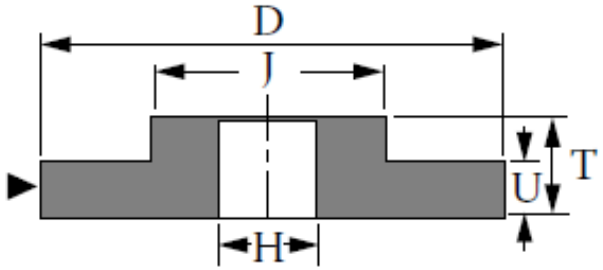
Relieved one side

Type 21



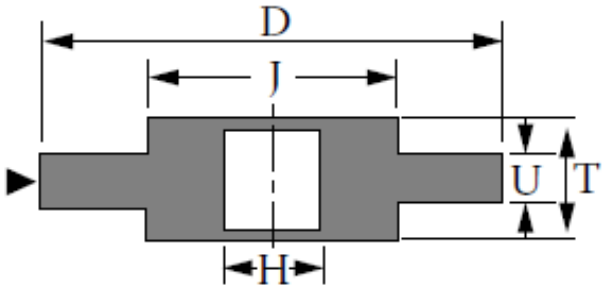
Relieved two sides

Type 38



Raised hub, one side

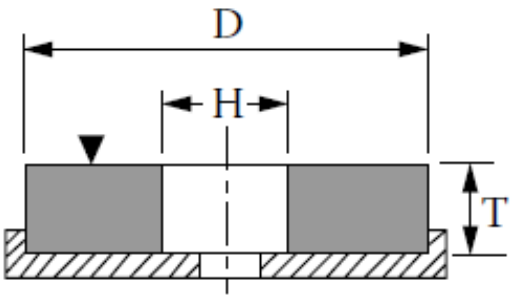
Type 39



Raised hub, both sides

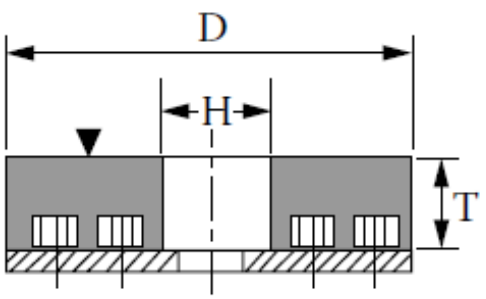
International Standard Shapes for Conventional Wheels

Type 35



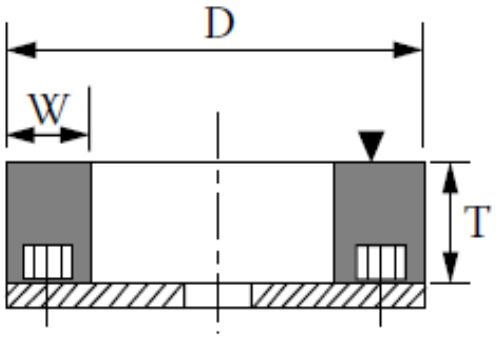
Plated disc

1 AN



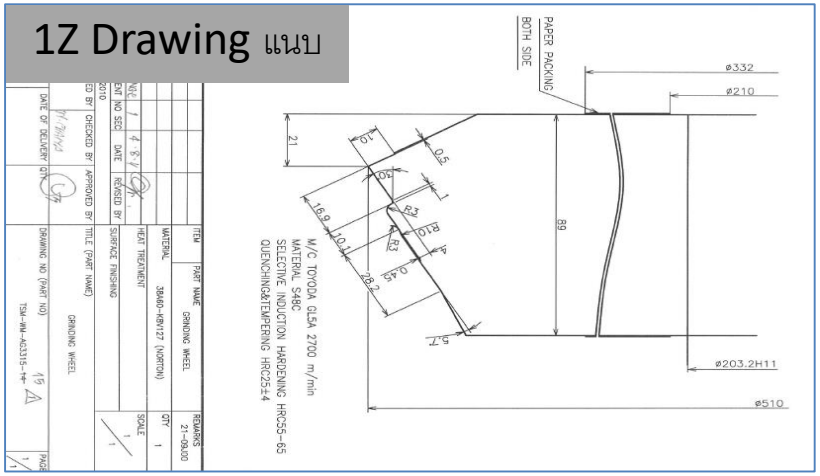
Inserted nut disc

2 AN

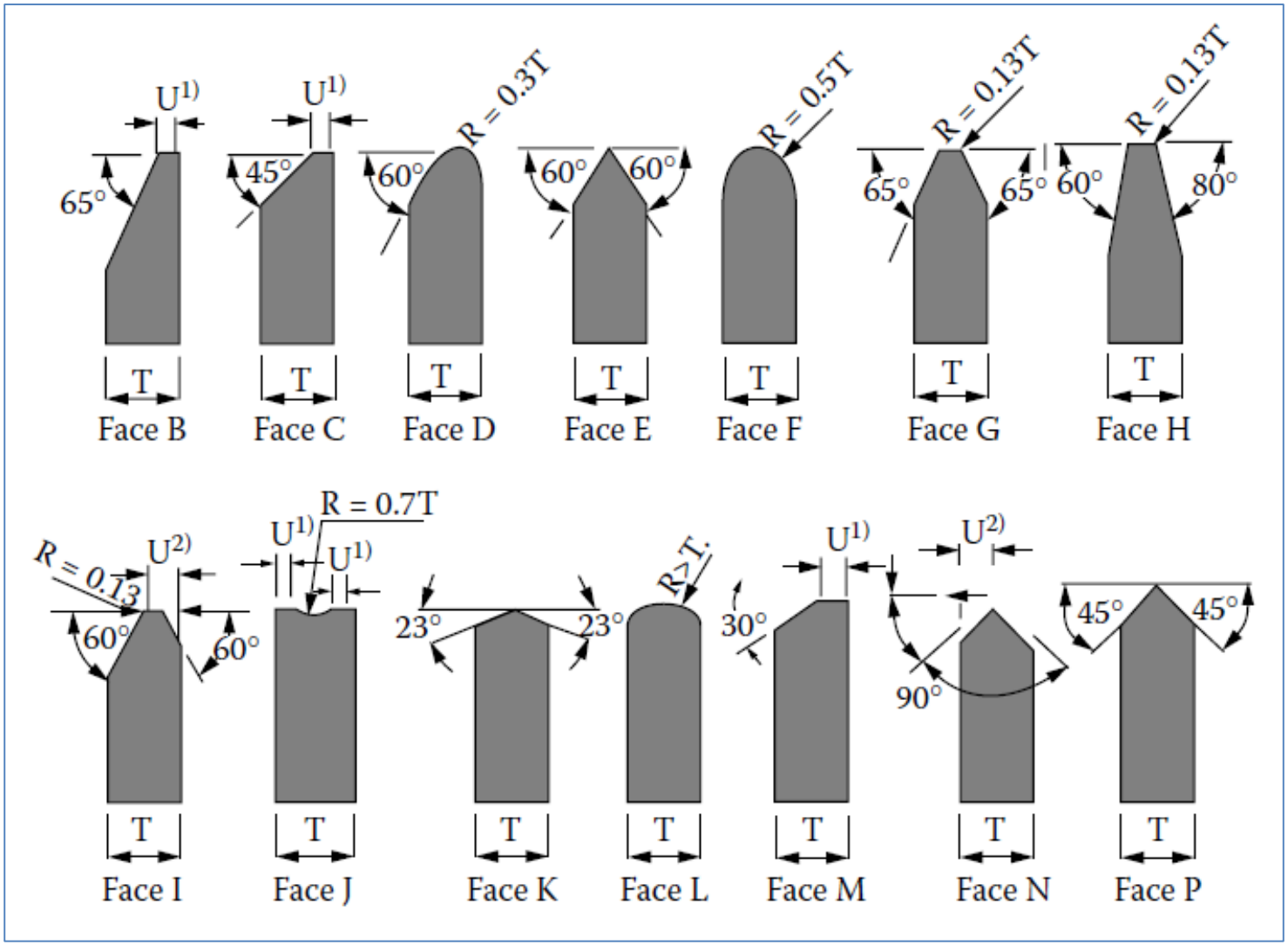


Inserted nut ring

1Z Drawing แนบ



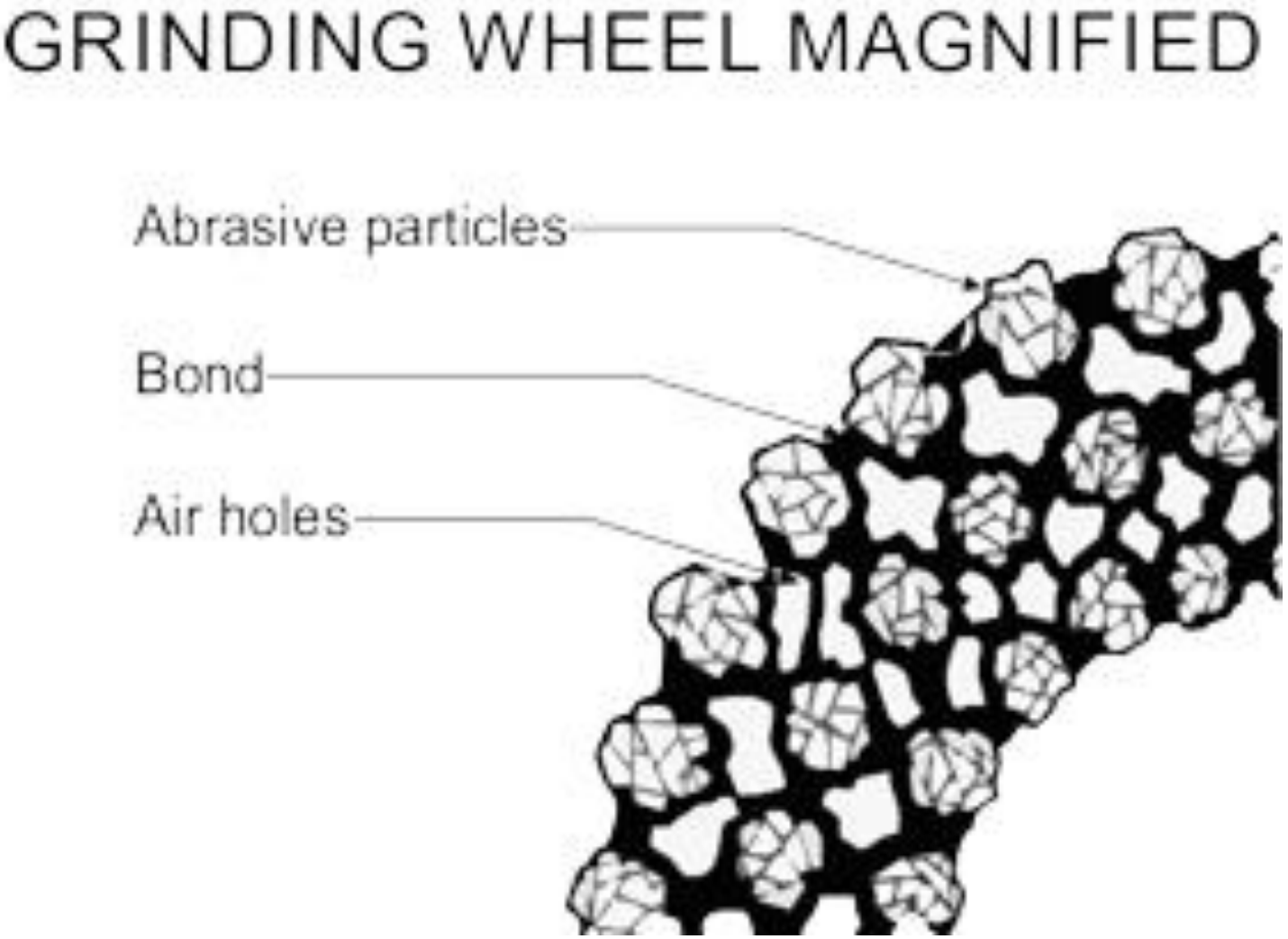
PROFILE (FACE)



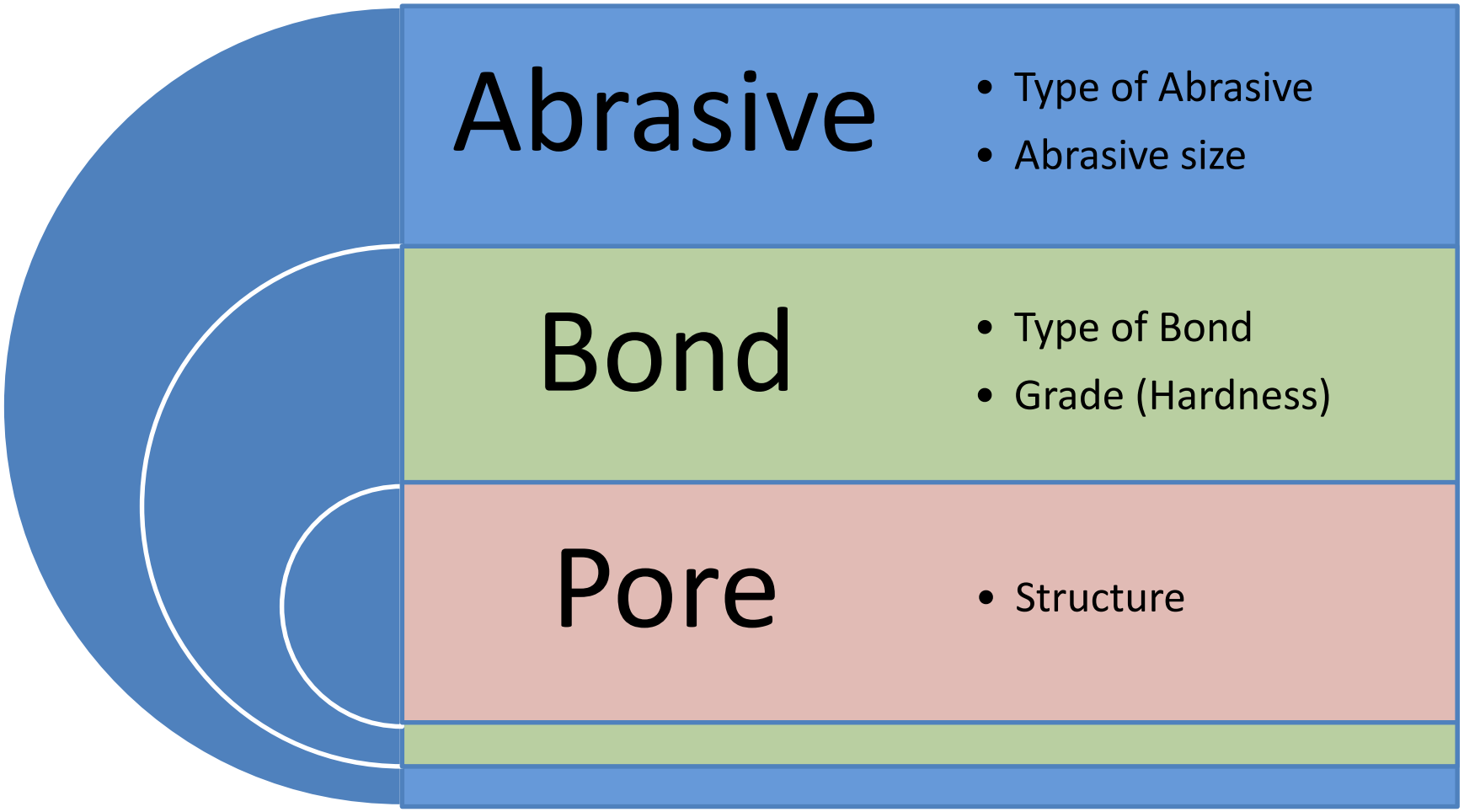
Chapter 5 The Nature of the Abrasive

- Introduction
- Alumina (Alox)-Based Abrasives
- Electrofused Alumina Abrasives
- Chemical Precipitation and/or Sintering of Alumina
- Silicon Carbide
- Diamond Abrasives
- CBN
- Grain Size Distributions

ส่วนประกอบของหินเจียร



3 Element & 5 Factors in Grinding wheel



Type of Abrasive

Alumina Oxide

- Regular
- Semi / Special
- Ceramic /Seed Gel /Cubitron

Silicon Carbide

- Black Silicon
- Green Silicon

ALUMINA (ALOX)-BASED ABRASIVES

- **Development of ALOX**

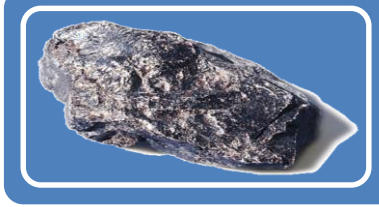
  Alumina-based abrasives are derived either from a traditional route of electrofusion, or more recently by chemical precipitation and/or sintering.

- **Manufacture**

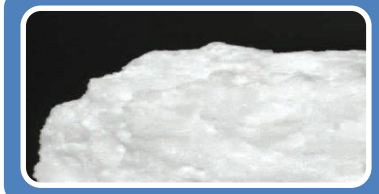
  raw material for electrofused alumina is bauxite(แร่อลูมิเนียม), contains 85 to 90% alumina, 2 to 5% TiO2, and up to 10% of iron oxide, silica, and basic oxides. The bauxite is fused in an electric-arc furnace at 2,600°C

Bauxite + TiO_2 + iron oxide, silica, basic oxides

Type of Abrasive : Aluminum Oxide



“A” The most widely used for heavy duty work such snagging steel casting, and for stock removal in Cylindrical grinding ,on all but the hardest and most heat sensitive steels like low alloy, Cast Steel and rough grinding application



“WA” More friable than “A” this also cool cutting grain. Most used for light grinding of all kind of hard , heat sensitive. It is excellent for tool room grinding sharpening of HSS, cast alloy tools like Hardened steel,tool steel SS (400series)



“PA” Very sharp and less friable than “WA” .Very cool cutting,retains better form and sharp cutting edge for a long time. Used for mounted point wheel , tool steel ,HSS.

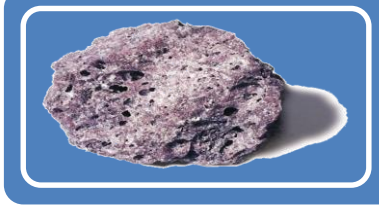


“EDR” Free cutting properties, slightly tougher and less friable than “WA” . Excellent for dry grinding in tool sharpening and tool room grinding application. Very cool cutting and sharp on 5% to 10% cobalt steels, Alloy HSS

Type of Abrasive : Aluminum Oxide



“SA” High precision grinding operation in V-bond (HRC>60) very hard steel in Surface grinding, tool room, gear grinding ,roll grinding



“ZP” Very good for profile holding , life time. Used in Surface grinding, Segment for valve ,steel plate.



“ZA” is a modified high zirconia used for foundries for snagging gray iron ,cast iron , malleable and ductile iron on floorstands and portable machine



“SL” Ceramic grain,The combination of the grains’ extreme sharpness and microstructure allows for an aggressive cutting and long-lasting ceramic grain ideal for low-pressure tough-to-grind applications. Very durable ,this abrasive is ideal for very high material removal.

SILICON CARBIDE

- **Development of SiC**

  SiC was first synthesized in 1891 by Dr. E. G. Acheson, who gave it the trade name “Carborundum.” It was initially produced in only small quantities as a substitute for diamond powder for lapping precious stones.

- **Manufacture of SiC**

  reaction of silica sand and coke at a temperature of around 2,400°C

$$SiO2 + 3C \rightarrow SiC + 2CO$$

Type of Abrasive : Silicon Carbide



“GC” Hard and friable , Used for grinding cement carbide tools hard and high chilled cast iron roll, non- metal , glass ,ceramics

Re grinding – truing in diamond tool / wheels



“C” very hard and friable than Aluminum Oxide, used for general grinding ,heavy duty snagging.

Hardness of Alumina and SiC

TABLE 5.1
Mechanical Properties of Typical Alumina and SiC Abrasives

Abrasive	Hardness Knoop	Relative Toughness	Shape/Morphology	Applications
Green SiC	2840	1.60	Sharp/angular/glassy	Carbide/ceramics/precision
Black SiC	2680	1.75	Sharp/angular/glassy	Cast iron/ceramics/ductile nonferrous metals
Ruby Alox	2260	1.55	Blocky/sharp-edged	Hss and high-alloy steel
White Alox	2120	1.75	Fractured facets/sharp	Precision ferrous
Brown Alox	2040	2.80	Blocky/faceted	General purpose
Alox/10% ZrO	1960	9.15	Blocky/rounded	Heavy-duty grinding
Alox/40% ZrO	1460	12.65	Blocky/rounded	Heavy-duty snagging
Sintered Alox	1370	15.40	Blocky/rounded/smooth	Foundry billets/ingots

GRAIN SIZE DISTRIBUTIONS

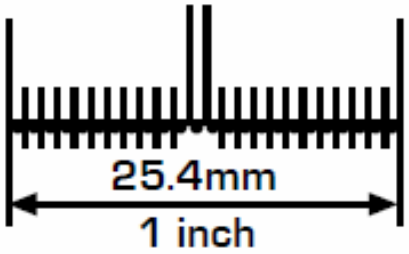
TABLE 5.3
Particle Size Comparisons

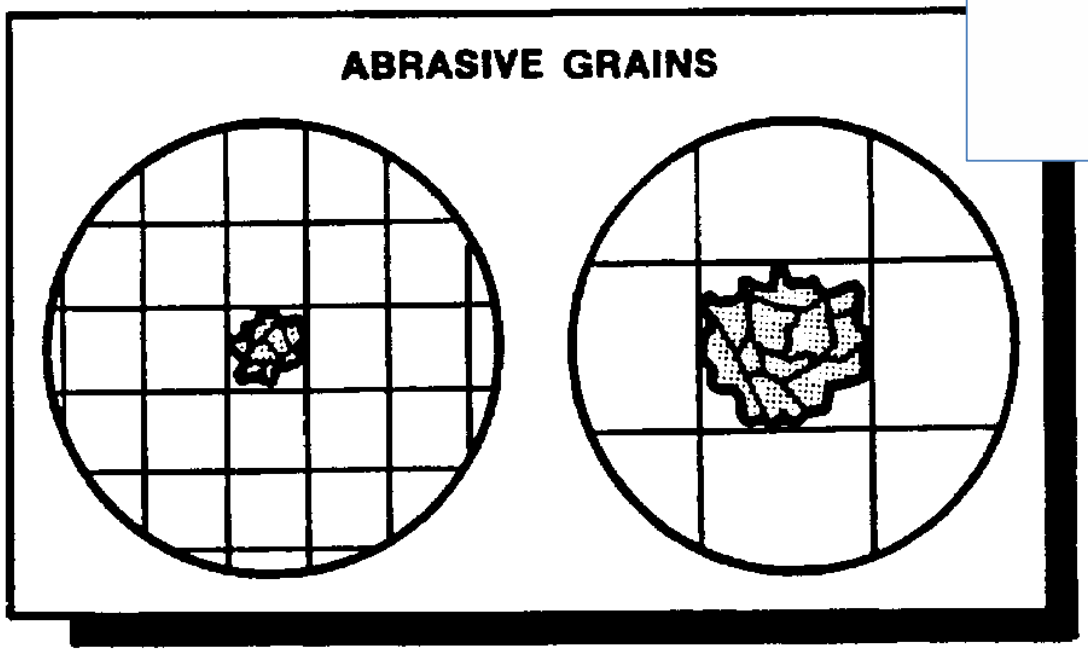
FEPA Designation	ISO R 5665—1990 Aperture Range (μm)	ANSI Grit Size	US Grit Number	Japanese (JIS) Size	Particles Per Carat
		Standard			
1181	1,180/1,000	16/18			
1001	1,000/850	18/20			
851	850/710	20/25			
711	710/600	25/30			
601	600/500	30/35			
501	500/425	35/40		35	
426	425/355	40/45			
356	355/300	45/50			
301	300/250	50/60	50	50	2,000
251	250/212	60/70	60		
213	212/180	70/80	80		
181	180/150	80/100	100	80	10,000
151	150/125	100/120	120	100	17,000
126	125/106	120/140	150	120	21,000
107	106/90	140/170	180	140	49,000
91	90/75	170/200	220	170	88,000
76	75/63	200/230	240	200	140,000
64	63/53	230/270	280	230	250,000
54	53/45	270/325	320	270	280,000
46	45/38	325/400	400	325	660,000

- THE ANSI STANDARD
  - Mesh size is defined by a pair of numbers that corresponds to sieves with particular mesh sizes.
- THE FEPA STANDARD
  - FEPA gives the grit size in microns of the larger mesh hole size through which the grit will just pass.
- US GRIT SIZE NUMBER
  - US grit size number with a single number that does not quite correlate with either the upper or lower ANSI grit size number.
- COMPARISON OF FEPA AND ANSI STANDARDS
  - FEPA has a tighter limit for oversize and undersize (5 to 12%) but no medium nominal particle size.
  - ANSI has somewhat more open limits for oversize and undersize (8 to 15%) but a targeted midpoint grit dimension.

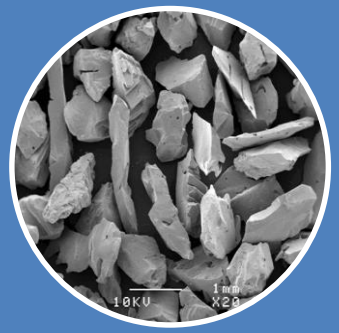
US GRIT NUMBER

$$1\text{mm} \approx = \frac{25.4}{24} = \text{grit 24 MESH}$$



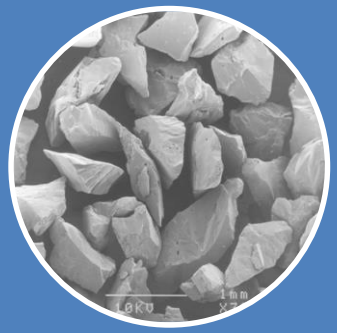


GRAIN SHAPE #60



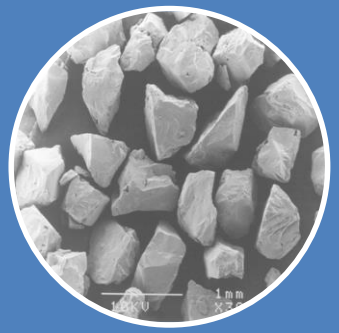
Point

Density : 1.57-1.67 g/cm3



Angular

Density : 1.67-1.77 g/cm3



Cubic

Density : 1.77-1.87 g/cm3

Chapter 6 Specification of the Bond

"POSTS" Holding abrasive grain in the wheel , the type of bond which depend on the wheel operating speed. The relative strength of the bond which holds the abrasive grains in place is called grade In general, with a given type of bond it is the amount of bond which determines the wheel's hardness. When the amount of bond is increased the size of the bond posts connecting each abrasive grain to its neighbors is increased. The larger bond post is naturally stronger, Thereby increasing the wheels hardness. The harder the wheel the more pressure and KWT(H.P) required to cause it to work efficiently

TYPE OF BONDING

Vitrified : Ceramic bond (V-bond)

- Work at Max 35 -48 m/s
- Porosity and strength of wheel made with this bond give high stock removal
- High precision work
- Not affected by water acid ,oil or ordinary temp variation

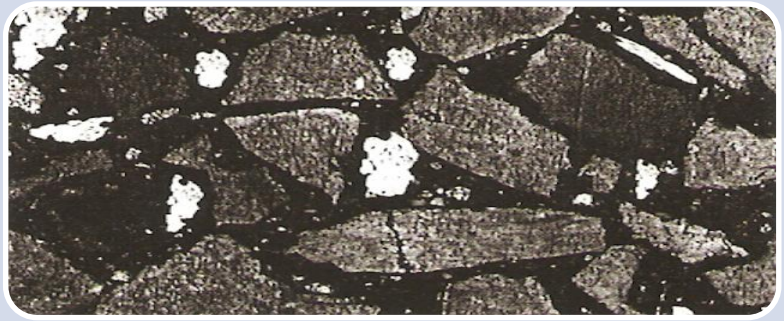
Resinoid : Synthetic Bond (B-bond)

- High Speed wheel 45-50m/s
- Excellent for rapid stock removal, heavy work.
- Affected by alkalis, humidity or extremes of climatic condition
- Deteriorated over a period

Strength of Bonding / Grade



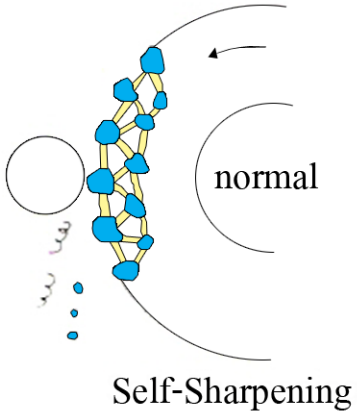
Abrasive grain of same size but with heavier bond coating and thicker , stronger bond posts as in a hard grade

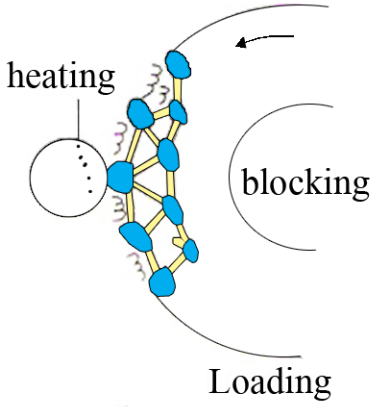


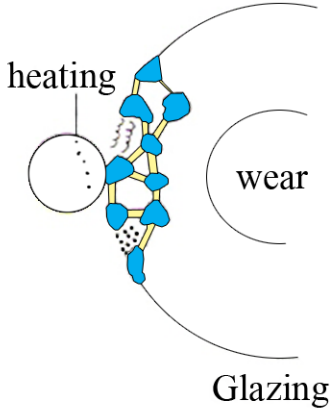
Abrasive grains with light bond coating and weak connecting bond posts as in a relatively soft grade wheel

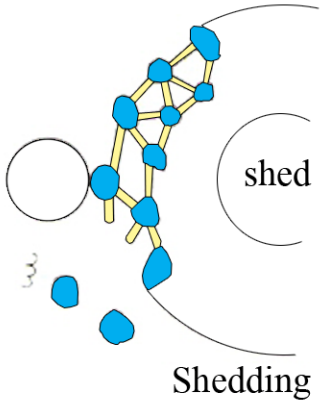
Bond - Grade Table

Very soft	soft	Medium	Hard	Very Hard
E,F,G	H,I,J	K,L,M,N	O,P,Q,R	S,T,U









Bond - Grade Table

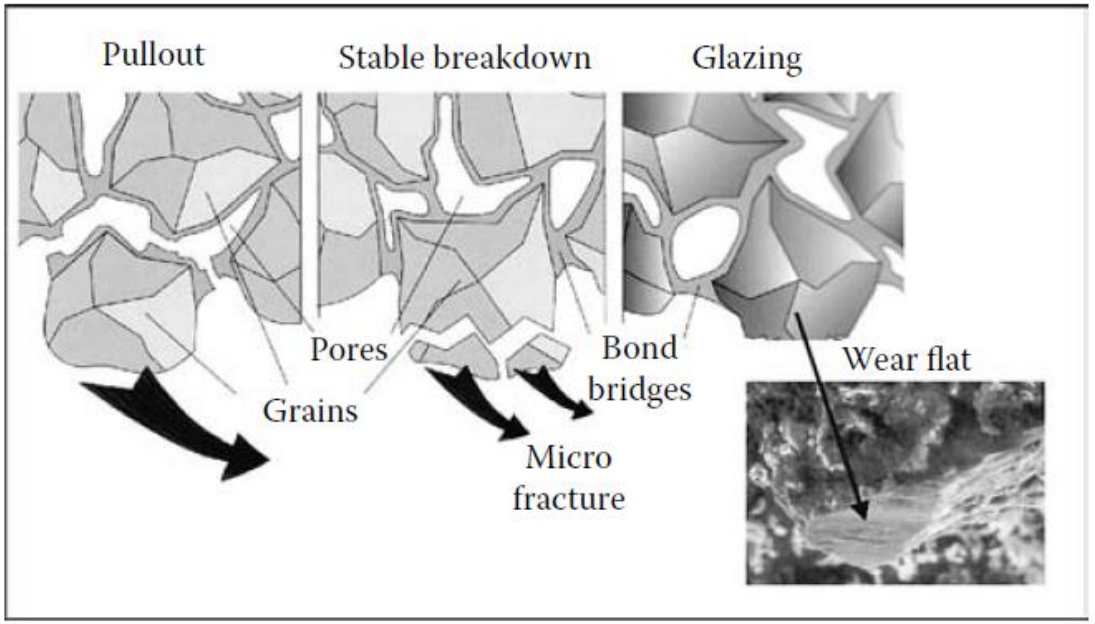
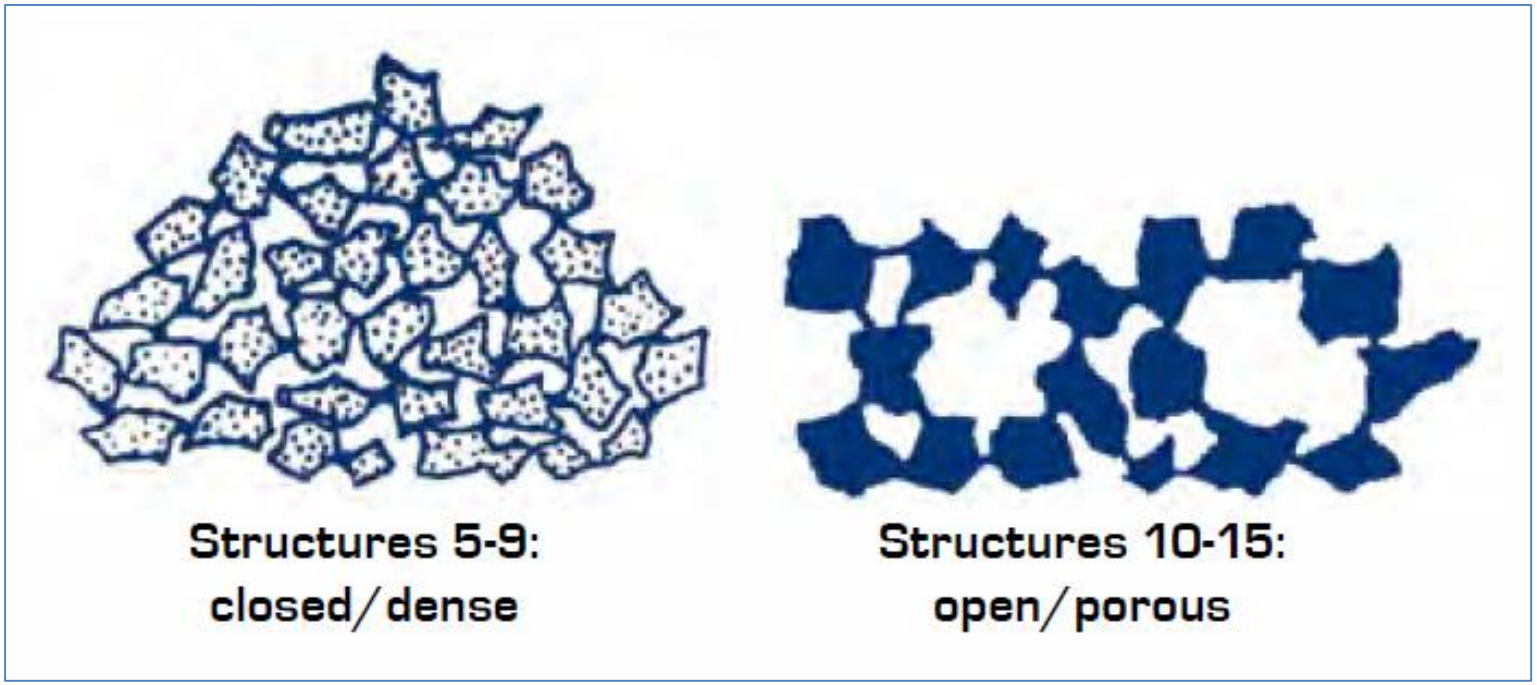


FIGURE 6.7 Pullout, stable breakdown, and glazing regimes in grinding. (Based on a drawing by Rappold 2002. With permission.)

Pore / Structure

The relative grain spacing is wheel, Dense space is denoted by low numbers and Open spacing by high numbers. Wheel structures are depending upon the material to be ground , rate of stock removal, accuracy and surface finish required.



Structure

Structure number	0	1	2	3	4	5	6	7	8
Abrasive volume percent	68	64	60	58	56	54	52	50	48

Kinik [n.d.] reported for their brand of wheels that structure number is related to abrasive volume by:

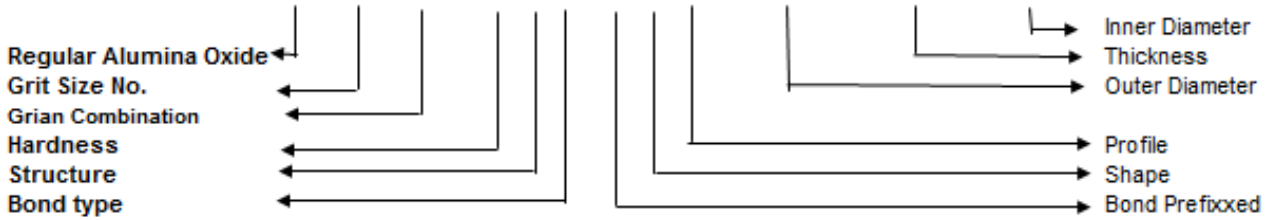
Structure number	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Abrasive volume percent	62	60	58	56	54	52	50	48	46	44	42	40	38	36	34

Grinding wheel Tracking system

Grinding wheel : Tracking system 2009

Position	1	2	3	4	5	6	7	8	9	10	11	12	13
	Abrasive type	**Grit No.**	**Grain combinatior**	**Hardness**	**Structure**	**Bond**	**Type of bonc**	**Filler type**	**Type**	**Profile**	**D: (mm**	**T (mm)**	**H (mm)**
	A: Rugula Brown	16	0.2	F-H :Very soft	4-6	V	01	S : Artificial Porosity	1	A	Outer D	Thickness	Inner D
	WA:White Alu	24	0.3	H-I :Soft	7-9	B	11	EL : Special Artificial	2	B			
	PA: Pink Alu	30	0.5	I-K :Medium	10-12	BF	12		3	C			
	BA:Blue Alu	36		L-N : Hard	14-17		15		4	D			
	SA: Monocrystalline	46		O-S : Very Hard			16		5	F			
	SL:321A	54					02		6				
	SP: Special Pink	60					03		7				
	19A: Low Density Alu	80					04		9				
	57A : High Density Alu	100					05		11				
	38A :Orange White	120					10		12				
	64A: Semi -	150							515				
	C:Black Silicon Carbide	180											
	GC: Green Silicon	220											
		240											
		320											
		400											
		500											
		600											

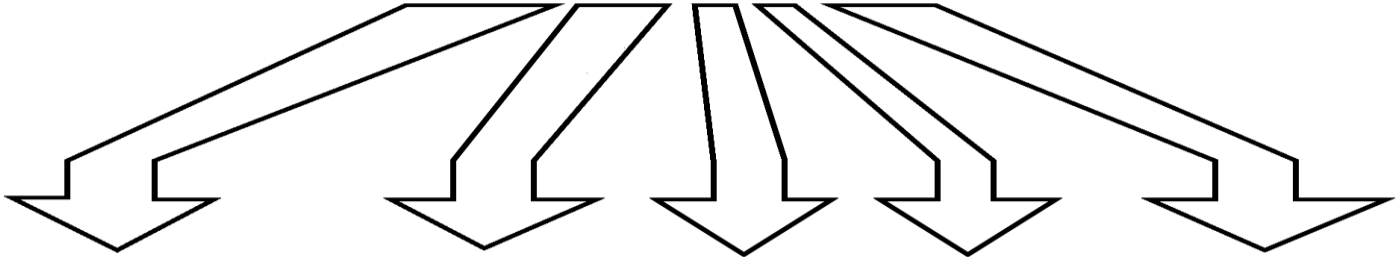
A 36 0.5 M 9 V 01 1 F 203.2 *19.05 *31.75



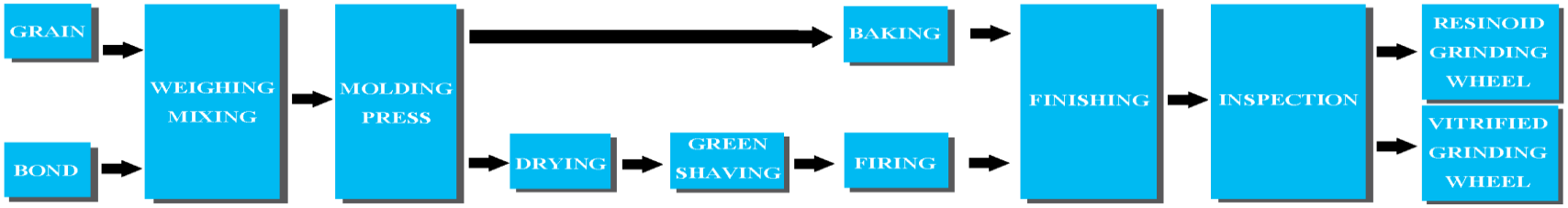
Specification and Manufacturing Process

Grinding Wheel Marking Summary

W A 4 6 J 7 B P

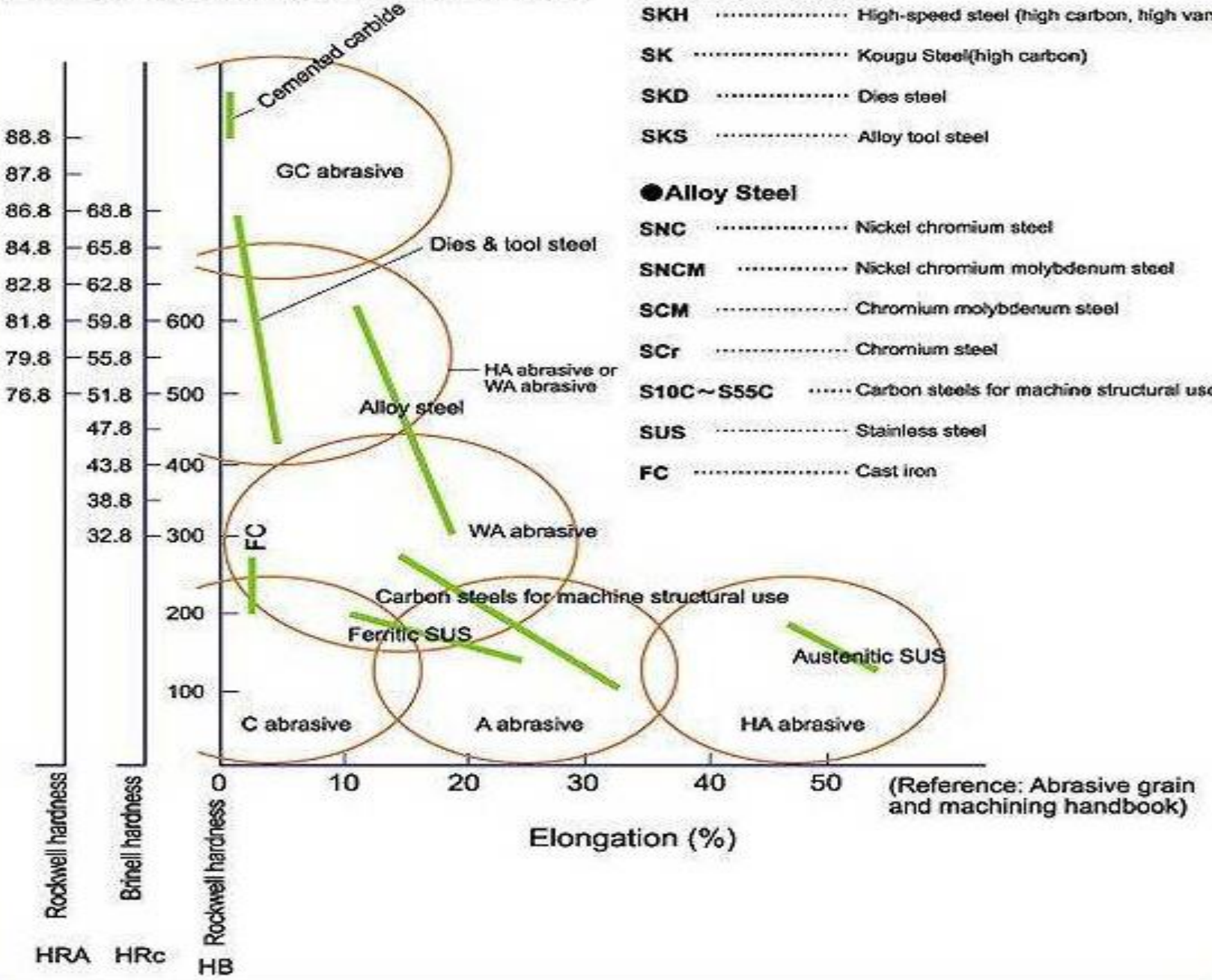


Abrasive	Grain size	Grade	Structure	Type of Bonds
Aluminium Oxide = A 64 Aluminium Oxide = 64A PA Aluminium Oxide = PA ZP Aluminium Oxide = ZP SA Aluminium Oxide = S A WA Aluminium Oxide = WA 73SA Aluminium Oxide = 73SA 64AF Aluminium Oxide = 64AF ZF Aluminium Oxide = ZF C Silicon Carbide = C GC Silicon Carbide = GC GC/C Silicon Carbide = GC/C A & C Mixture = AC	Coarse To Very Fine 10 30 70 220 12 36 80 240 14 46 90 280 16 54 100 320 20 60 120 400 24 150 500 180 600	Soft To Very Hard E J O U F K P V G L Q W H M R X I N S Y T Z	1 13 2 14 3 15 4 16 5 17 6 18 7 19 8 20 9 21 10 22 11 23 12 24	B = RESINOID ALUMINIUM OXIDE AND SILICON CARBIDE PRODUCT B = Regular Resinoid BP = Regular "P" Type Induced Pore V = VITRIFIED ALUMINIUM OXIDE PRODUCT V = Regular Vitrified VBEP = "BEP" Type Vitrified Pore Vitrified



Conversion spec of SL Carborundum and Competitor

SL	MGT	Noritake	Norton	Tyrolit	Winter tool	Kinik
A	A	A,GA	A,LA	10A,52A	10A	A
WA,38A	WA	WA	WA,38A	89A	42A,53A	WA,38A
PA	PA	PA,RA	86A	88A	57A	PA,RA
SA	H	SA,SN	32A,SA	90A	67A	PSA,32A
64A	19A	19A,81A	19A,57A	69A	31A	FA,57A
SL	MC	CX	SGB			
3SL	3MC	CXY	3SG			
5SL	5MC	CXZ	5SG			
ZA		ZF,NZ	ZF,NZ			AZ
C	C	C	37C	37C	C	C
GC	GC	GC	39C	39C	C	GC
GC/C	BGC	C/GC	C/GC	50C		
70AC	37A	A/C	A/C			


[Hardness & Elongation of Workpiece and Selection of Abrasive]
●Dies & Tool Steel
SKH ····· High-speed steel (high carbon, high vanadium)
SK ····· Kougu Steel(high carbon)
SKD ····· Dies steel
SKS ····· Alloy tool steel
●Alloy Steel
SNC ····· Nickel chromium steel
SNCM ····· Nickel chromium molybdenum steel
SCM ····· Chromium molybdenum steel
SCr ····· Chromium steel
S10C~S55C ····· Carbon steels for machine structural use
SUS ····· Stainless steel
FC ····· Cast iron
Cemented carbide
GC abrasive
Dies & tool steel
HA abrasive or WA abrasive
Alloy steel
WA abrasive
FC
Carbon steels for machine structural use
Ferritic SUS
Austenitic SUS
C abrasive
A abrasive
HA abrasive
88.8
87.8
86.8
84.8
82.8
81.8
79.8
76.8
68.8
65.8
62.8
59.8
55.8
51.8
47.8
43.8
38.8
32.8
600
500
400
300
200
100
0
10
20
30
40
50
Rockwell hardness
HRA
Brinell hardness
HRc
Rockwell hardness
HB
Elongation (%)
(Reference: Abrasive grain and machining handbook)